人工智能原理与应用

第三章：自然语言处理

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个人简介

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主要发表地： ICLR、ICML、 NeurIPS、 SIGIR、KDD、WWW等

科研成果：发表录用长文30余篇，其中超过5篇论文入选了最高引和最具影响力榜单，Google Scholar引用5700余次，h-index 30，获DASFAA 2026最佳论文奖，入选斯坦福“全球前2%顶尖科学家”

个人主页：https://wujcan.github.io/

提纲

1 语言表示与词向量

2 序列模型与注意力机制

3 Transformer与预训练模型

4 上下文工程

5 大语言模型训练

6 大模型智能体

ChatGPT, Gemini, Qwen … 是语言模型

Prompt： 请根据鲁迅《孔乙己》原文继续一下文字：孔乙己一到店，所有喝酒的人便都看着他笑

DeepSeek：

好的，我们来续写这段经典开头。

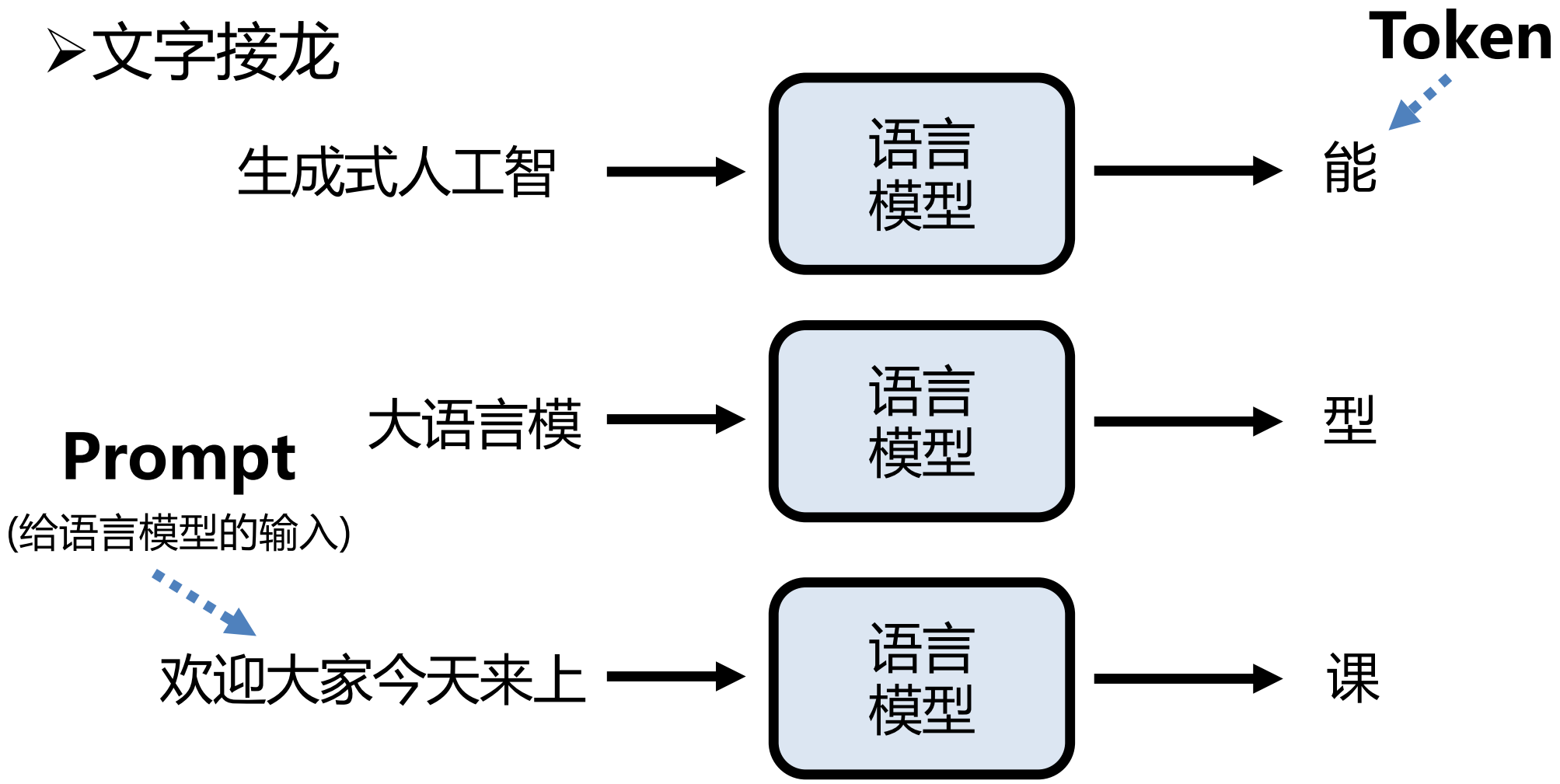
孔乙己一到店，所有喝酒的人便都看着他笑。有的叫道，“孔乙己，你脸上又添上新伤疤了！”他不回答，对柜里说，“温两碗酒，要一碟茴香豆。”便排出九文大钱。他们又故意的高声嚷道，“你一定又偷了人家的东西了！”孔乙己睁大眼睛说，“你怎么这样凭空污人清白……”“什么清白？我前天亲眼见你偷了何家的书，吊着打。”孔乙己便涨红了脸，额上的青筋条条绽出，争辩道，“窃书不能算偷……窃书！……读书人的事，能算偷么？”接连便是难懂的话，什么“君子固穷”，什么“者乎”之类，引得众人都哄笑起来：店内外充满了快活的空气。

Ground Truth

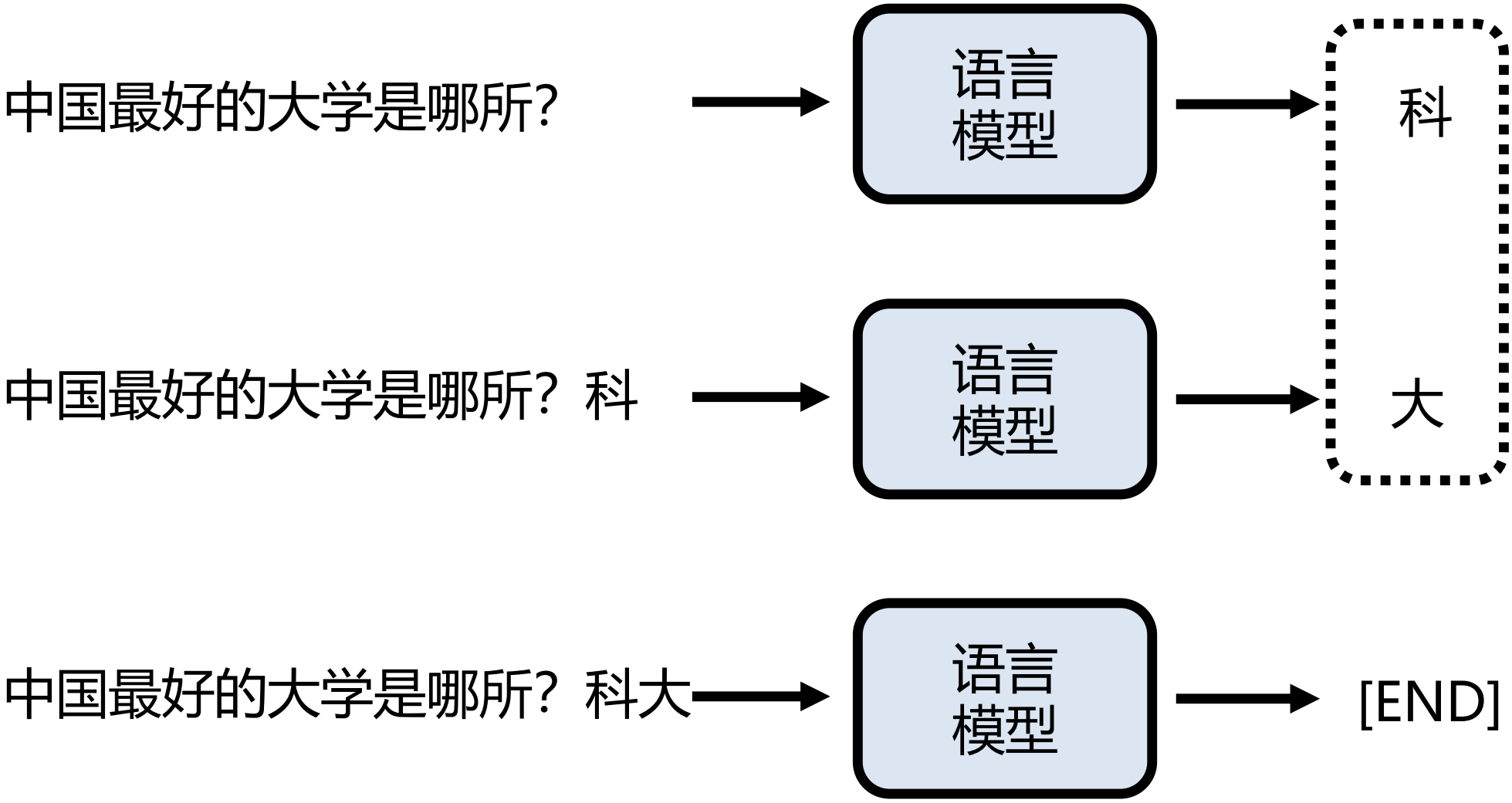
孔，别人便从描红纸上的“上大人孔乙己[7]”这半懂不懂的话里，替他取下一个绰号，叫作孔乙己。孔乙己一到店，所有喝酒的人便都看着他笑，有的叫道，“孔乙己，你脸上又添上新伤疤了！”他不回答，对柜里说，“温两碗酒，要一碟茴香豆。”便排出九文大钱。他们又故意的高声嚷道，“你一定又偷了人家的东西了！”孔乙己睁大眼睛说，“你怎么这样凭空污人清白……”“什么清白？我前天亲眼见你偷了何家的书，吊着打。”孔乙己便涨红了脸，额上的青筋条条绽出，争辩道，“窃书不能算偷……窃书！……读书人的事，能算偷么？”接连便是难懂的话，什么“君子固穷[8]”，什么“者乎”之类，引得众人都哄笑起来：店内外充满了快活的空气。

ChatGPT, Gemini, Qwen … 是语言模型

- 文字接龙

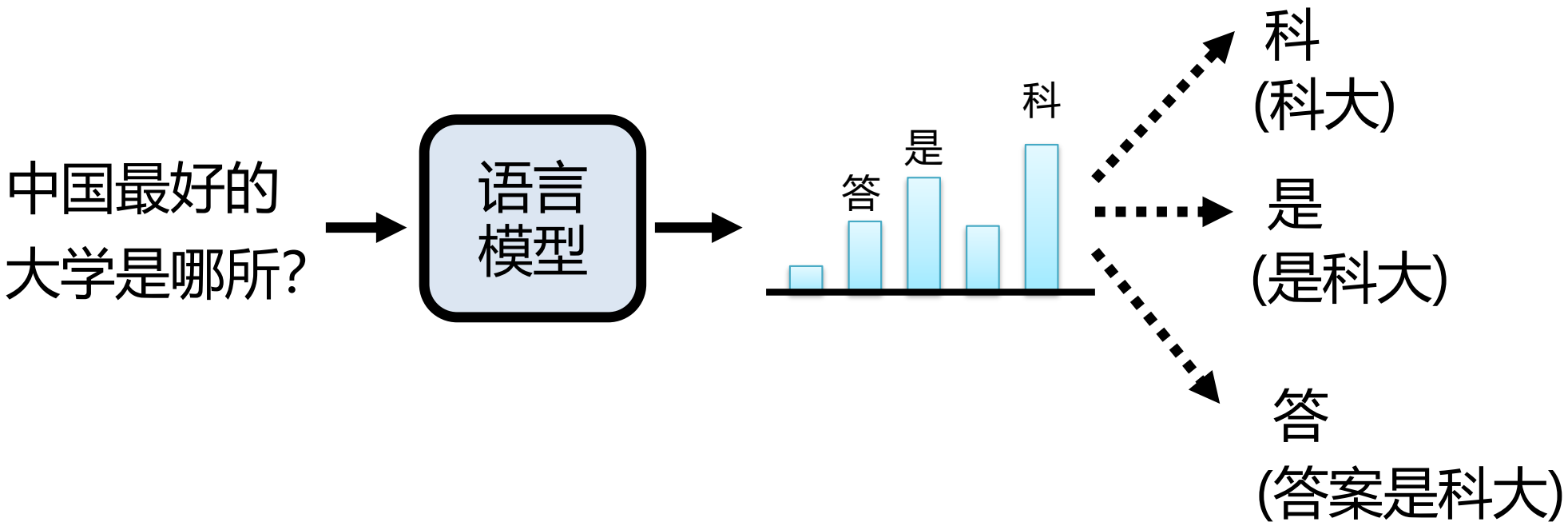


语言模型怎么回应呢?



下一个 token 的概率 = P(下一个 token | 前面已经出现的所有 token)

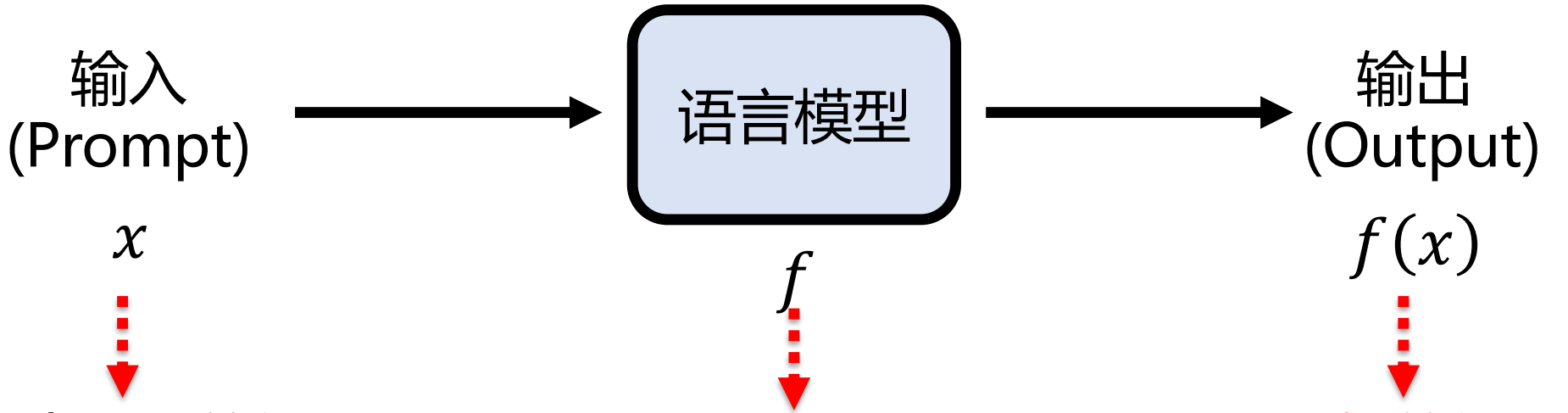
语言模型每次的回答都可能不同



但是你不用担心模型可能会莫名其妙地回答“薯条”，
因为“中国最好的大学是哪所？ “后面接薯的概率非常低。

语言模型的运作原理?

内容关系



- 更为合适的输入
  - 语言表示与词向量
  - 上下文工程
- 模型结构原理
  - 序列模型
  - 注意力机制
  - Transformer
- 怎么学到的?
  - 预训练
  - 后训练

更好的输出

词的表示

计算机如何理解人类语言?

人类语言是离散的符号系统

Level 1 · 字

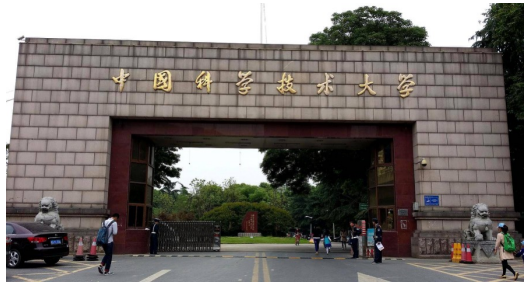
猫　狗　鱼　鸟

Level 2 · 词

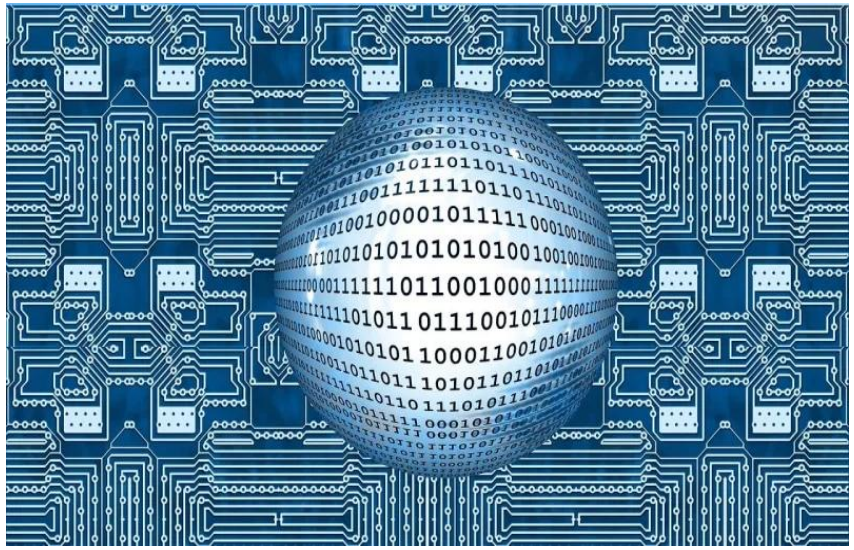
自然语言 | 机器学习 | 深度学习

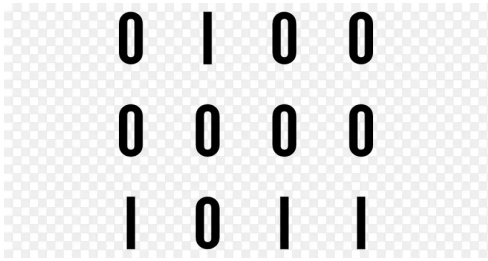
Level 3 · 句子

中国最好的大学是科大。



计算机只能处理0和1





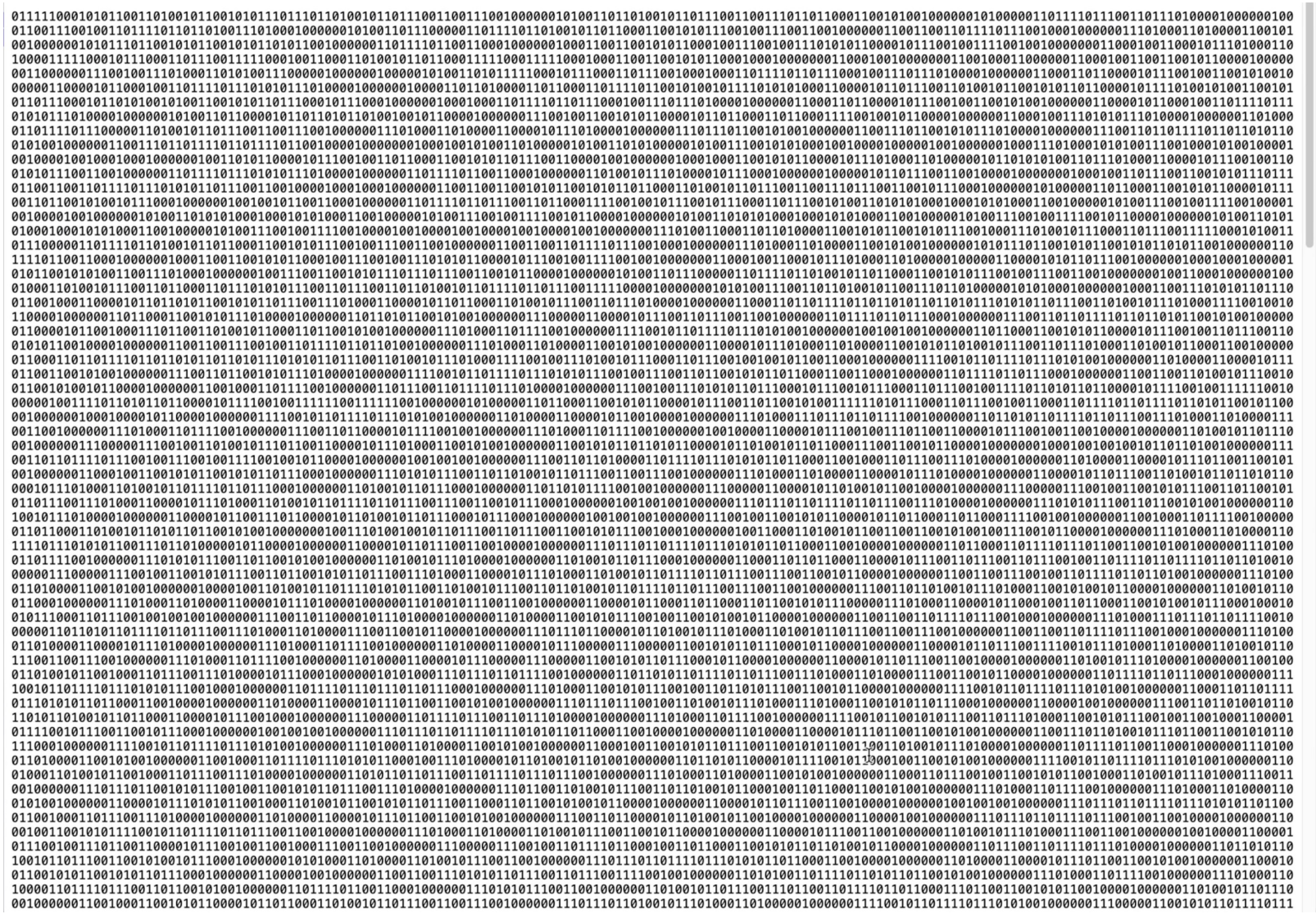
需要一种“翻译”机制

词的表示

|Viewing Single Post From: Spoilers for the Week of February 11th|\n|Lil||Feb 1 2013, 09:58 AM|\nDon't care about Chloe/Taniel/Jen-Jen. Don't care about Sami, really, but hoping that we get some good "SAMANTHA GENE!!" Marlena Death-Stares out of it. And "newfound" feelings. Please. If only.\nSTEFANO!! STEFANO, STEFANO, STEFANO!!!! :cheer:\n|Spoilers for the Week of February 11th · DAYS: News, Spoilers & Discussion| *sigh* Fundamentalist community, let me pass on some advice to you I learned from the atheistic community:\nIf you have set yourself on fire, do not run.\nOkay? Okay?? Please?\nLook, D, you had two months to say to Harvard in private emails, "Im sorry, I shouldnt have been using that animation in my paid presentations. I wont use it again. I really do like 'Inner Life', though, and would love to use it in classroom presentations, from the BioVisions site, if that is acceptable."\nI sat here, for two months, waiting for that to happen, anything to happen, and it didnt. Two months, on your own terms, you could have written a similar post to yesterdays. I would have given you the benefit of the doubt-- maybe you didnt know the credits werent visible to the audience, and I wouldnt have said a word beyond this, as its Harvards problem, not mine. This would have been a funny joke to those of us involved in dealing with you people, but it would have been a PR non-issue for you.\nBut after you set yourself on fire, you didnt douse it out with a bucket of ice cold reality and accountability. You ran. And youre still running.\nWhy not just state "I screwed up. Sorry everyone." and move on? Why the excuses? Why the denial? Why the passive language? Why the vague words and cryptic capitalizations? Why the writes and rewrites of your 'press release'? We know it wasnt written of your own volition, or it would have been done *before* Harvard had to take action. And, your behavior before this, regarding this issue, is not indicative of someone who made an innocent mistake. Its weird.\nSo what with this frantic running? Is the inability to say "I was wrong" a pathological feature of Creationists? Or are you hiding something? Or is it both? Or is it more?\nAnd now we get Casey weighing in on the issue, according to cre8id at AboveTopSecret.com-- PBS/NOVA online - Intelligent Design on trial:\n...to my knowledge, Discovery Institute has neither authorized nor received nor is making use of any presentation that used that animation. We have had nothing to do with creating or selling a DVD of that animation, nor do we have anything to do with placing that presentation on Google Video.I dont know what he is talking about with that last part, but the first part sounds similar to DIs claims post-Dover ("WE HAD NOTHING TO DO WITH DOVER!"). Maybe Luskin is telling the truth. Maybe this was a magic non-science Creation-friendly narration with convenient edits that AiG or ICR would have killed for... but only Dembski could find it... but he cant tell us where... and he didnt share it with anyone... and its subsequently disappeared from the Internet...\nBut that simply isnt what Ive been told. Maybe this was all a silly Dembski mistake, blown out of proportion due to his decision to remain silent... But what if we find more videos of more DI fellows, presenting this animation?\n*shrug* If youve set yourself on fire, do not run, DI. If youve done it, better admit it to Harvard now, apologize, and move on. A novel two-step immunotherapy approach has shown clinically beneficial responses in patients with advanced ovarian cancer. Following Lifestyle Recommendations Reduces Risk of Cancer Death\nPeople who follow the diet and lifestyle recommendations laid out by the WCRF and the AICR have a 20 percent reduced risk of dying from cancer. UCSF Launches Social Networking Site for Patients and Families with Hereditary Cancers\nFor Immediate Release May 14, 2013 UCSF Launches Social Networking Site for Patients and Families... Genomic Test May Help Guide Prostate Cancer Treatment\nThe Oncotype DX® Prostate Cancer Test strongly predicts aggressiveness of disease. Statins Linked to Lower Risk of Liver Cancer in Hepatitis C\nPeople infected with chronic hepatitis C are less likely to develop liver cancer if they are taking statins.\nRadioimmunotherapy (RIT) is a type of targeted therapy that delivers radiation directly to cancer cells.... Urinary Incontinence\nOverview The urinary tract includes the kidneys, the ureters, the bladder, and the urethra. The kidneys... Advanced Directives\nLiving Wills Every competent adult has, in most cases, the freedom to accept or refuse medical treatment.... Caregivers\nWhat is Caregiving and Who are Caregivers? Caregivers are individuals who provide care to chronically... Chemotherapy for Older Patients: What You Should Know About the Risk of Infection\nAs you may already know, chemotherapy works by attacking the rapidly dividing cells it finds in the body,...\nAn ongoing series highlighting complementary therapies, adapted from The Complete Guide to Complementary... Clear and precise\nMohs surgery provides a tissue-sparing approach to skin cancer surgery. By Eleanor Mayfield Michele Kelsey... Chemical Reaction\nChemicals may be disrupting our hormones-and our health. By Laurie Wertich Exposure to synthetic chemicals... College Kids Kick Cancer\nBy Diana Price College kids and cancer-not two topics most of us would immediately connect. And yet... Cooking with Fruits and Vegetables\nIn the introduction to Ripe: A Fresh, Colorful Approach to Fruits and Vegetables (Running Press, 2011;...\nAnnual meeting brings together cancer experts from around the world. Kari Bohlke, ScD The 2011 Annual... Bone Fractures in Breast Cancer Patients More Frequent with Femara than with Tamoxifen\nResearchers affiliated with the BIG I-98 Collaborative and International Breast Study Groups... Single Treatment with High-intensity Focused Ultrasound Effective for Localized Prostate Cancer\nResearchers from McMaster University in Canada have reported that high-intensity focused... Marital Separation Impacts Cancer Survival\nResearchers from the University of Indiana and the Fox Chase Cancer Center... 2009 Oncology Conference Coverage View up-to-date coverage of the 2009 Oncology Conference here. Free the Cans! Working Together to Reduce Waste\nIn a blog about how people share, it's worth the occasional reference to the bizarre ways that people DON'T SHARE. Is it safe to say we live in a society that places great value on independence, private property, personal space, and privacy? Even sometimes extreme value? Is that why people at an 8-unit apartment building in Oakland, CA have separate caged stalls for eight separate trash cans? I know it's not nice to stare, but I walked by these incarcerated cans and could not help myself. I returned with my camera, so that I could share my question with the world: Why can't people share trash cans or a single dumpster? Or, at the very least, why can't the cans share driveway space?\nThe Zero Waste Movement has come to the Bay Area and it calls for a new use for these eight cages. Here are my suggestions:\n- Turn two of those cages into compost bins. Fill one with grass, leaves, and vegetable scraps, let it decompose for six months, then start filling the second bin in the meantime.\n- Put in a green can, which is what Oakland uses to collect milk cartons, pizza boxes, yard trimmings, and all food to send it to the municipal composting facility. If your city doesn't do this yet, tell them it's a great idea and they could be as cool and cutting edge as Oakland.\n- Put in one or two recycling cans for glass, plastic, cardboard, paper, aluminum, etc.\n- Put out a FREE STUFF box for unwanted clothing and household items. The neighbors could sort through it each week, and later put it out on the curb for passers-by to explore. Take what's left to Goodwill or a comparable donation spot.\n- Put in a few small bins for various items that can be recycled, such asbatteries and electronics, which can then be taken to an electronics recycling center every month or two. Styrofoam can be brought to a local packaging store or ceramics business that accepts used packaging material. Or, if you accumulate a bunch of plastic bags,take them to a store or to some other place that accepts used ones.\n- Put in ONE trash can. By the time you compost, recycle, re-use, redistribute, and take a few other measures to reduce your waste, you'll have almost no trash each week.\n- Install a bicycle rack or locked bicycle cage.\n- With the leftover space, put in a container garden and a bench where neighbors can gather and chat. A much more pleasant alternative to the garbage can jailhouse ambiance, wouldn't you agree? ORLANDO, Fla. — While the Rapid Recall Exchange, the 2-year-old industry recall portal, has signed up more than 600 manufacturer and retailer subscribers, it still lacks the "critical mass" of suppliers that would make it a primary source of recall information, according to trade association officials and retailers.\nManufacturers use the exchange to communicate timely and accurate product recall and withdrawal notifications to retailer and wholesaler headquarters, which in turn share the information with individual stores. The exchange's retail membership represents 85% of U.S. grocery volume – including 21 of the 24 largest supermarket chains based in the United States – but it still lacks key suppliers, especially in the fresh food sectors, said Pat Walsh, senior vice president, industry relations, education and research for Food Marketing Institute, Arlington, Va.\n"We have good penetration [among manufacturers] on the dry grocery side – though it needs to be better – and need to expand in other fresh food verticals like meat, produce, deli and bakery," said Walsh, who participated in a session on the RRE at the U Connect Live conference here earlier this month.\nMajor food distributors in the exchange, including Kroger, Wegmans and Wakefern, have recently sent letters to their vendors explaining that the only way they will accept recall information is via the RRE, noted Brian Lynch, senior director of business and industry development for the Grocery Manufacturers Association, Washington, who also participated in the U Connect Live session. In an April letter posted on www.rapidrecallexchange.org, Kroger asked all of its suppliers to subscribe to the exchange by July 1.\nMichael Roberson, director of corporate quality assurance for Publix Super Markets, Lakeland, Fla., said in the U Connect Live session that the chain is "disappointed" in the number of manufacturers

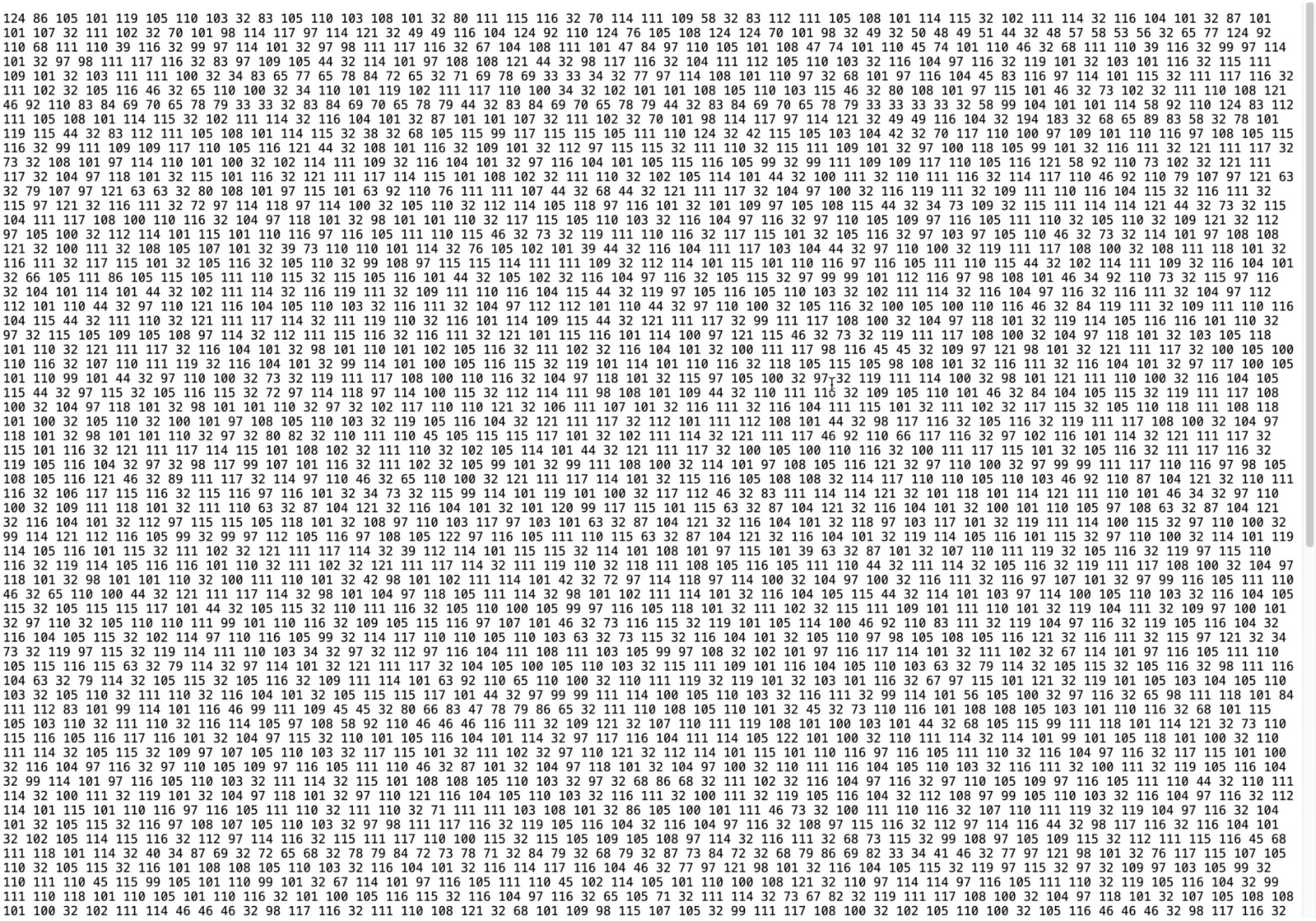
一维文本序列

词的表示



文本经过UTF-8编码成“0/1”比特（bit）

词的表示



每8个比特形成1个字节（Byte），共$256 = 2^8$个组合方式

词的表示



如果每个字节替换成独特的emoji表情

独热编码（One-hot Encoding）

或许我们可以不用给每个字符一个唯一编码。。。

给词典中每个词分配一个唯一编号，用一个稀疏高维向量表示

- 设表示词的字典为V，可以将字典中第i个词用维度为$|V|$的独热(One-Hot)向量进行表示：

$$w_i = [0, 0, \dots, 1, \dots, 0] \in \{0, 1\}^{|V|}$$

- 示例

词	编号	One-hot向量
猫	0	**1**,0,0,0
狗	1	0,**1**,0,0
鱼	2	0,0,**1**,0
鸟	3	0,0,0,**1**

特点：N个词 → N维向量，每次只有一位为1

为什么独热编码不够好?

- **维度灾难**

  词表有多大，向量就有多长： ~10万+词

- **无法表达语义相似性**

  任意两个词之间的距离都完全相等 $Cos_sim\left(\boldsymbol{v}_{猫}, \boldsymbol{v}_{狗}\right) = 0$

- **信息密度极低**

  信息利用率趋近于零，对神经网络的权重矩阵乘法来说是极大的浪费

- **词序与上下文（Context）被忽略**

  词在句子中的位置信息？上下文词语的关系？词的多义性？

- **无法处理未知词（Out of Vocabulary）**

  词表固定后，训练集之外的词无法被表示→ 丢弃或者< UNK >替代

我们希望：低维度 + 有语义 + 能计算

分布式语义假设

➢ 分布假设

"You shall know a word by the company it keeps."

— J. R. Firth, 1957 · 英国语言学家

➢ 句子示例

“我养了一只猫，它_____”

很可爱　　毛茸茸的　　喜欢喵喵叫

这些上下文词 → 帮助定义"猫"的含义

方法：潜在语义分析

- 基于矩阵分解获得词分布表示
  - 记词为w，上下文为c，二者共现矩阵为M
  - 上下文c的选择：
    - 固定窗口的词：反映词法、句法等局部属性
    - 所在文档：反映词代表的主题信息

	c_1	c_2	c_3	...	$c_{\|C\|}$	$count(w)$
w_1	2	8	12	...	125	570
w_2	0	0	60	...	9	168
w_3	1004	987	19	...	0	3089
...	...	...	...	...	...	...
$w_{\|V\|}$	0	19	2039	...	2	5760
$count(c)$	1997	2010	4323	...	239	213985

- 点互信息 $PMI(w,c) = \log_2 \frac{p(w,c)}{p(w)p(c)}$，其中 $p(w) = \frac{count(w)}{count(all)}$

点互信息PMI

➢ PMI 能够消除“巧合高频”偏差

事件	出现天数	边际概率
下雨	20 天	P(雨) = 0.2
打伞	21 天（基本只在雨天打）	P(伞) = 0.21
吃饭	100 天（天天吃）	P(饭) = 1.0
下雨 ∩ 打伞	20 天	P(雨,伞) = 0.20
下雨 ∩ 吃饭	20 天	P(雨,饭) = 0.20

① 下雨 & 打伞

$$\mathrm{PMI}(雨, 伞) = \log \frac{0.20}{0.2 \times 0.21} = \log \frac{0.20}{0.042} = \log 4.76 \approx +2.25$$

- **PMI 远大于 0**，说明两者有强烈的真实关联

② 下雨 & 吃饭

$$\mathrm{PMI}(雨, 饭) = \log \frac{0.20}{0.2 \times 1.0} = \log \frac{0.20}{0.20} = \log 1 = 0$$

- **PMI 等于 0**，共现完全可以被”两者各自的高频”解释，没有额外关联

方法：潜在语义分析

- 对于处理过的共现矩阵M，利用截断奇异值分解获取词表示：

$$M \approx U\Sigma \mathrm{V}^{\top}$$

$$U \in \mathbb{R}^{|V| \times d}, V \in \mathbb{R}^{d \times |C|}$$为正交矩阵

其中，U的每一行代表每个词的d维向量表示，U的各列之间正交（词表示每一维度正交）

- 词表示的每一维度表达了一种独立的**潜在语义**
- 类似的$\Sigma \mathrm{V}^{\top}$的每一列也可以作为向量表示

举例：用”甜度、辣度、咸度”这3个维度来描述所有食物：
糖果 → [0.9, 0.0, 0.1]
辣椒 → [0.0, 1.0, 0.2]
薯片 → [0.1, 0.3, 0.9]

方法：基于神经网络的词分布表示

- 从离散到连续表征：
  - 将词的one-hot编码通过一个神经网络映射到一个d维的连续向量空间，每个词对应空间中一个点
  - 利用自然语言文本的自监督学习信号（即词与上下文的共现信息）进行词向量训练
- 经典模型Word2vec：
  - Word2vec本质是只有一个隐藏层的神经网络，学习两种映射：one-hot → word embedding → one-hot
  - 按照训练目标（语义关联方式）的不同分为两种模型CBOW和Skip-gram

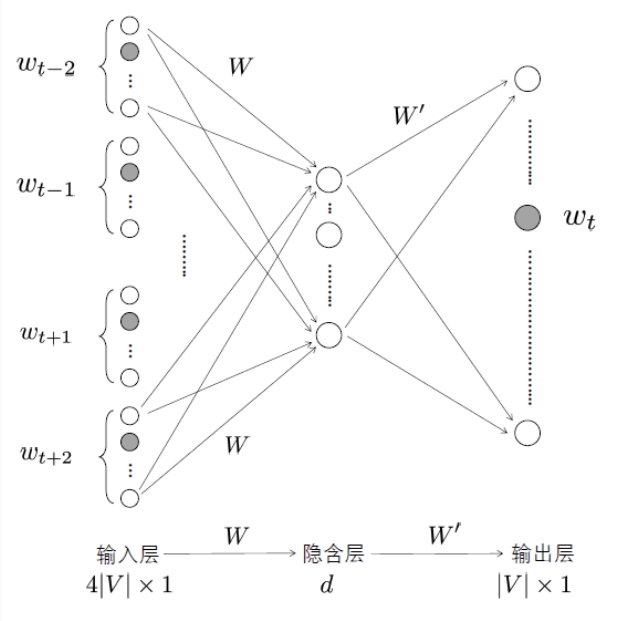
Cbow vs Skip-gram

CBOW
Continuous Bag of Words

VS.

Skip-gram

类比：完形填空

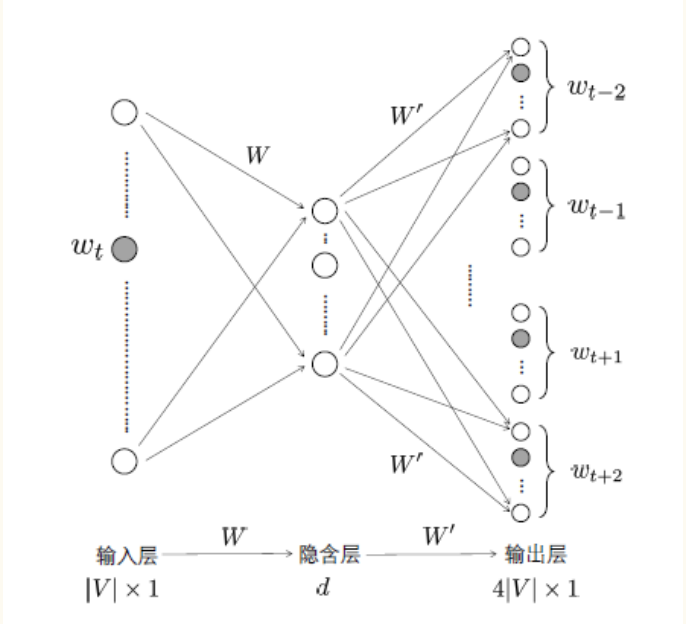


[我] [喜欢] [可爱]

↓

预测中心词"猫"

类比：造句练习

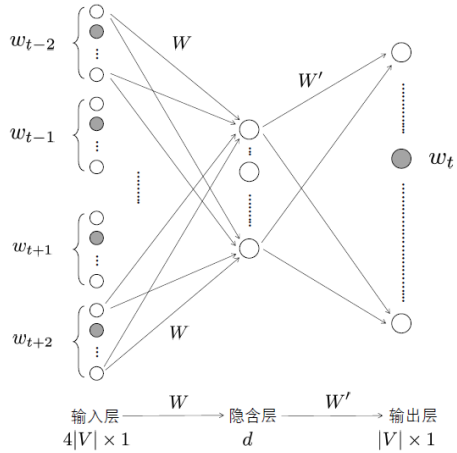


中心词 [猫]

↓

预测 [我] [喜欢] [可爱]

Cbow模型架构



- 输入层
  - 记上下文单词的独热表示向量$OneHot(w_i)$
- 隐含层
  - 首先将输入层每个单词利用矩阵$W \in \mathrm{R}^{d \times |V|}$映射至隐空间
  - 设 $v(w_i) = \boldsymbol{W} \cdot OneHot(w_i)$
  - 对于多个词组成的上下文C，对所有的词向量取**平均值**作为上下文表示

$$v(C) = \frac{1}{|C|} \sum_{\boldsymbol{w} \in \boldsymbol{C}} v(w)$$

- 输出层
  - 令$W' \in \mathrm{R}^{|V| \times d}$作为输出层权值矩阵。利用$W'$和$W$相应行列内积进行结果概率预测
  - 记W'的隐含空间表示为$v'(w)$

$$p(w_t|C) = \frac{e^{<v(C), v'(w_t)>}}{\sum_{w' \in V} e^{<v(C), v'(w')>}}$$

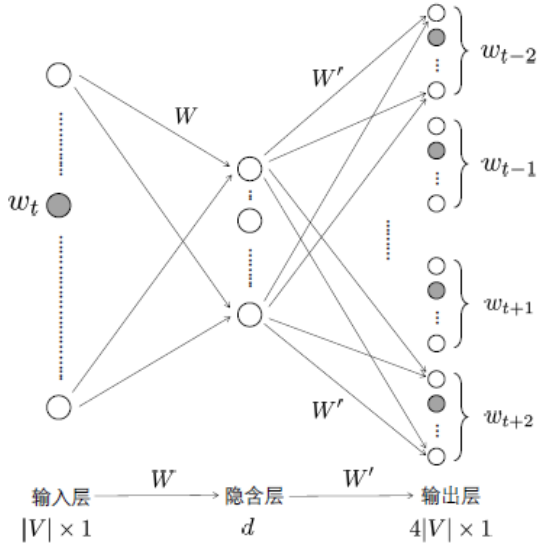
- 最终W或W'或二者组合均可作为词向量矩阵

Skip-gram模型

- CBOW模型利用上下文窗口中的词$C = \{w_{t-k}, \dots, w_{t-1}, w_{t+1}, \dots, w_{t+k}\}$计算$p(w_t \mid C)$

- Skip-gram对C进行了简化，每个词都可以做独立的上下文，计算概率改变为 $p(w_t|w_{t+\mathrm{j}})\ (j \in \pm 1, \dots \pm k)$
  - 原论文采用了等价描述$p(w_{t+\mathrm{j}}|w_t)$, 计算方法为：

$$p(w_{t+j}|w_t) = \frac{e^{<v(w_t), v\prime(w_{t+\mathrm{j}})>}}{\sum_{w\prime \in V} e^{<v(w_t), v\prime(w\prime)>}}$$



Cbow 与 Skip-gram的参数优化

- 对于两种方法，都可以通过构造似然损失函数利用随机梯度下降法来最大似然估计W'和W
  - CBOW负对数似然损失函数：

$$\mathcal{L}(\theta) = -\sum_{t=1}^{T} \log p(w_t|C)$$

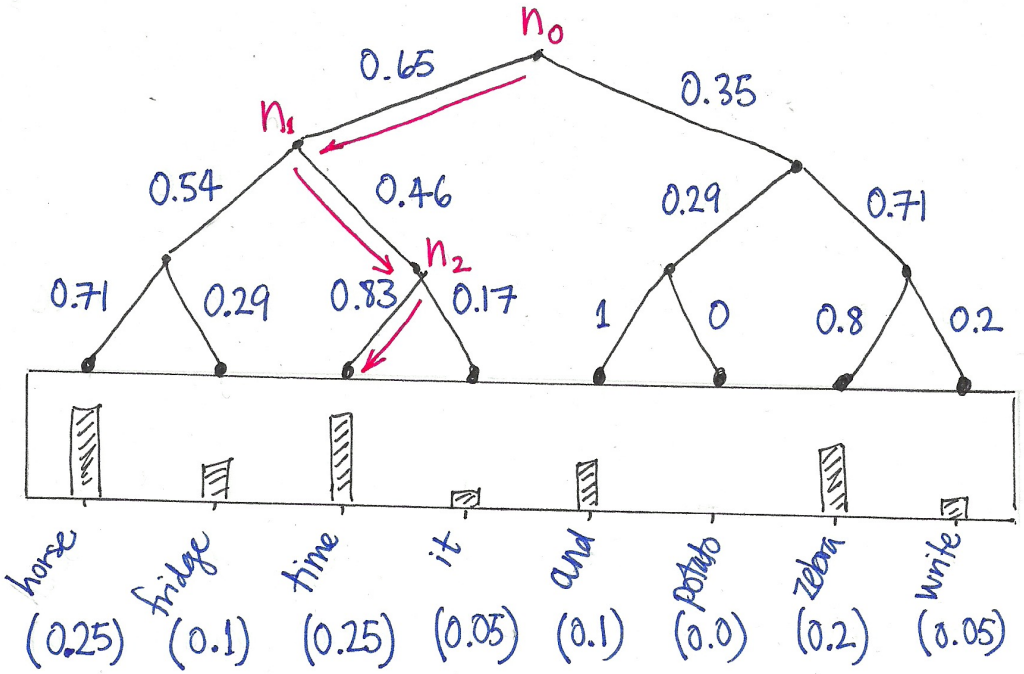
  - Skip-gram负对数似然损失函数：

$$\mathcal{L}(\theta) = -\sum_{t=1}^{T} \sum_{-k \le j \le k, j \ne 0} \log p(w_{t+j}|w_t)$$

  - 实际操作中，输出层softmax函数维度过大，运行效率特别慢
  - 解决效率问题的两种近似方法
    - 层次Softmax（Hierarchical Softmax）
    - 负采样方法（Negative Sampling）

层次Softmax（Hierarchical Softmax）

- 在 Word2Vec 中



- 每次更新参数都要

$P(\cdot \mid c)$

- **层次Softmax**：
  - 训练前根据词
  - 在每个内部节点 n，做一次二元逻辑回归，决定向左还是向右走

$$p(\text{左} \mid n) = \sigma\left(\boldsymbol{v}'_n{}^{\top}\boldsymbol{h}\right), \qquad p(\text{右} \mid n) = 1 - \sigma\left(\boldsymbol{v}'_n{}^{\top}\boldsymbol{h}\right)$$

  - **某个词 w 的概率 = 路径上所有节点概率之积:**

$$p(w \mid C) = \prod_{i=1}^{L(w)-1} \sigma\left(\text{sign}[n(w, i+1) = \text{左}] \cdot \boldsymbol{v}'_n{}^{\top}\boldsymbol{h}\right)$$

  - 复杂度为$O(\log_2|V|)$

https://building-babylon.net/2017/08/01/hierarchical-softmax/

基于负采样的词向量学习

- 进行任务修改
  - 原任务：从给定上下文预测目标单词
  - 新任务：给定上下文和目标单词，判断是否共现
    - 由$|V|$分类转化为2分类。

- 以Skip-gram为例
  - 对于每一对目标词和上下文w_t, w_{t+j}为正例
  - 取K个不在上下文窗口内的$\widetilde{w_i}$作为负例
  - 利用下式进行参数估计

$$\max \log\ \sigma\left(< v(w_t), v'(w_{t+j}) >\right) + \sum_{i=1}^{K} \sigma\left(-< v(w_t), v'(\widetilde{w_i}) >\right)$$

基于负采样的词向量学习

- 举例
  - 文本： 

  - 正例：
    - (brown, fox)
    - (brown, quick)
    - (brown, jumps)
  - 负例：
    - (brown, over)
    - (brown, lazy)
    - ……

从语料到词向量的训练全流程



Step 1

准备语料

维基百科
新闻文章
书籍等大规模文本



Step 2

滑动窗口

在文本上滑动提取
(中心词,上下文词)
配对样本



Step 3

神经网络训练

反复迭代
调整权重矩阵
最小化预测误差



Step 4

词向量

隐藏层权重
就是每个词的
稠密向量表示

训练完成后，每个词对应一个 100~300 维的稠密向量，像查字典一样可以直接使用

词表示评价

- 词义相关性
  - 人工标注一些词对的**相似度。**
  - 利用词向量**余弦相似度值**与人工标注相似度的**相关系数**作为词义相关性的评价标准：

Word 1	Word 2	Human (mean)
tiger	cat	7.35
tiger	tiger	10
book	paper	7.46
computer	internet	7.58
plane	car	5.77
professor	doctor	6.62
stock	phone	1.62
stock	CD	1.31
stock	jaguar	0.92

Dataset example: WordSim 353

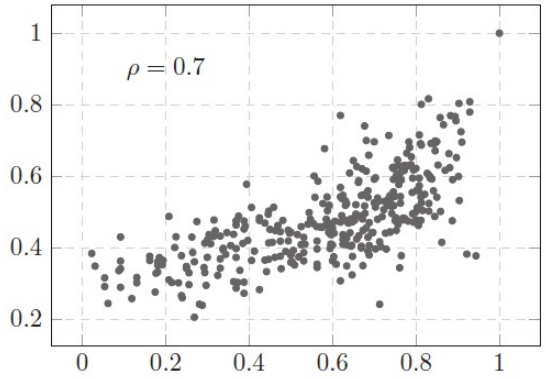


图 5.6 根据词向量计算得到的*cosine*相似度（纵轴）与WordSim353人工标注（横轴）之间的相关性。Pearson相关系数为0.7。

词表示评价

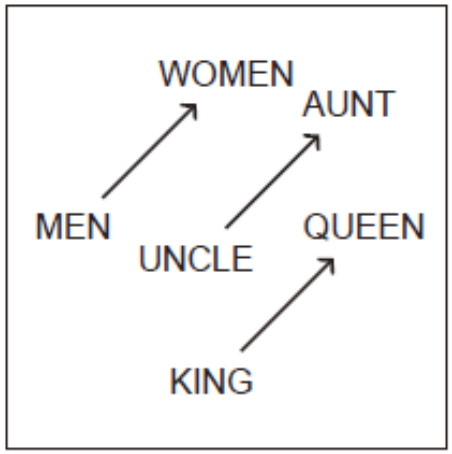
➢ 类比推理性

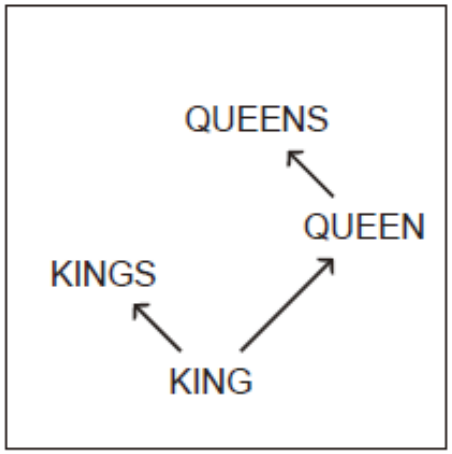
- 如果两个词对(w_a, w_b)和(w_c, w_d)之间存在词法或语义上的类比，那么存在：

$$v(w_b) - v(w_a) \approx v(w_d) - v(w_c)$$

- 如在下图例子中：

$$v(WOMEN) - v(MEN) \approx v(QUEEN) - v(KING)$$





提纲

1 语言表示与词向量

2 **序列模型与注意力机制**

3 Transformer与预训练模型

4 上下文工程

5 大语言模型后训练

语言的序列特性

- **词序决定语义**



VS



- **长距离依赖**

长距离关联

主语 谓语

The **cat,** which already ate all the fish, **was** full.

- **上下文决定语义**

我吃了一个苹果→ 🍎 水果 **VS** 他用苹果手机→ 📱 品牌

短语和句子表示

- 句子的表征（Representation）：
  - 句子的表征包含其中词语的关联语义信息，可以直接用于各种下游任务
  - 句子表征通过融合其中的词向量得到

- 融合方式
  - 词袋模型
  - 循环神经网络
  - 注意力模型

词袋模型

➢ 基本词袋模型

- 将文本看作是由词构成的集合，用维度为$|V|$的向量X作为文本表示，X_i代表第i个词是否出现在该文本中或者是在该文本中出现的次数

- 直观例子：假设语料库共 **3 个文档**：
  - 文档1：**"我 爱 学习 学习 很 重要"**
  - 文档2：**"我 爱 打球"**
  - 文档3：**"学习 打球 都 重要"**

词表（|V| = 7）：[我, 爱, 学习, 很, 重要, 打球, 都]

	我	爱	学习	很	重要	打球	都
文档1	1	1	**2**	1	1	0	0
文档2	1	1	0	0	0	1	0
文档3	0	0	1	0	1	1	1

词袋模型

- 基于词频的词袋模型
  - 第i个词出现在该文本中的频率
  - 一般需要归一化，防止文本长度带来的负面影响

$$\text{词频(TF)} = \frac{\text{某个词在文章中的出现次数}}{\text{文章的总词数}}$$

文档1的TF

词	出现次数	TF
我	1	$1/6 \approx 0.167$
爱	1	$1/6 \approx 0.167$
学习	**2**	$2/6 \approx 0.333$
很	1	$1/6 \approx 0.167$
重要	1	$1/6 \approx 0.167$
打球	0	0
都	0	0

$$v_{\text{文档}_1} = [0.167, 0.167, 0.333, 0.167, 0.167, 0, 0]$$

词袋模型

- 基于TF-IDF（词频-逆文件频率）的词袋模型
  - IDF：

$$\text{逆文档频率(IDF)} = \log\left(\frac{\text{语料库的文档总数}}{\text{包含该词的文档数} + 1}\right)$$

  - TF-IDF：字词的重要性随着它在文件中出现的次数成正比增加，但同时会随着它在语料库中出现的频率成反比下降。

$$\text{TF} - \text{IDF} = \text{词频(TF)} \times \text{逆文档频率（IDF）}$$

各词的IDF（共3个文档）

词	出现在几个文档	IDF
我	2	$\log(3/3) = 0$
爱	2	$\log(3/3) = 0$
学习	2	$\log(3/3) = 0$
很	1	$\log(3/2) \approx 0.405$
重要	2	$\log(3/3) = 0$
打球	2	$\log(3/3) = 0$
都	1	$\log(3/2) \approx 0.405$

文档1的TF-IDF

词	TF	IDF	TF-IDF
我	0.167	0	**0**
爱	0.167	0	**0**
学习	0.333	0	**0**
很	0.167	0.405	**0.068**
重要	0.167	0	**0**
打球	0	0	**0**
都	0	0.405	**0**

词袋模型

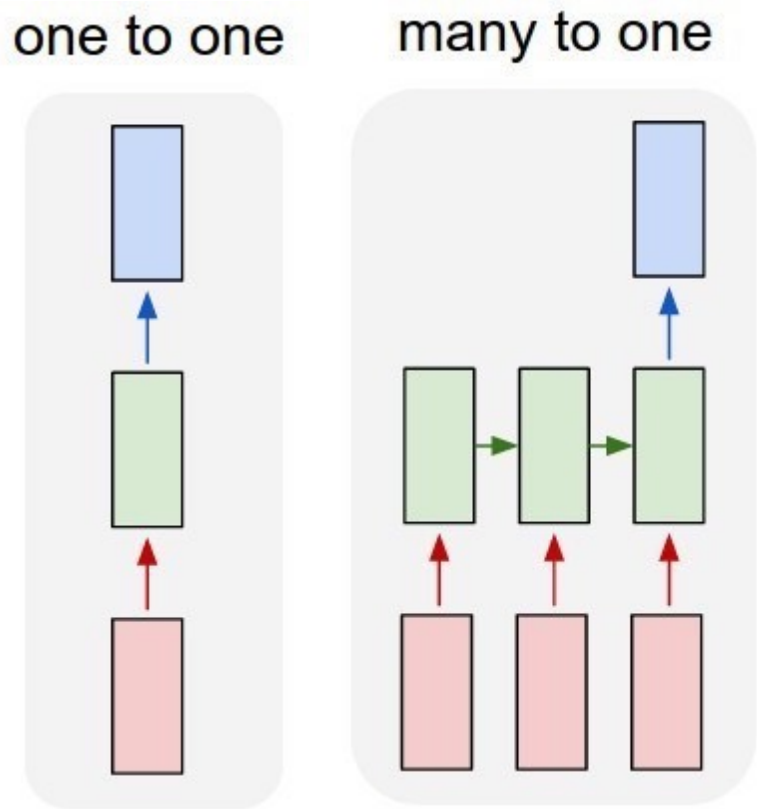
- 基于词的分布式表示的词袋模型
  - 对于任何分布式单词表示，句子或文本中包含的所有单词的分布式表示的**平均值**被用作为它们表示。

- 词袋模型的缺点
  - 没有注意到**语法**和**词序**对句子表示的影响

循环神经网络RNN

- 为什么传统的前馈网络和卷积网络不适合处理文本（序列）数据?

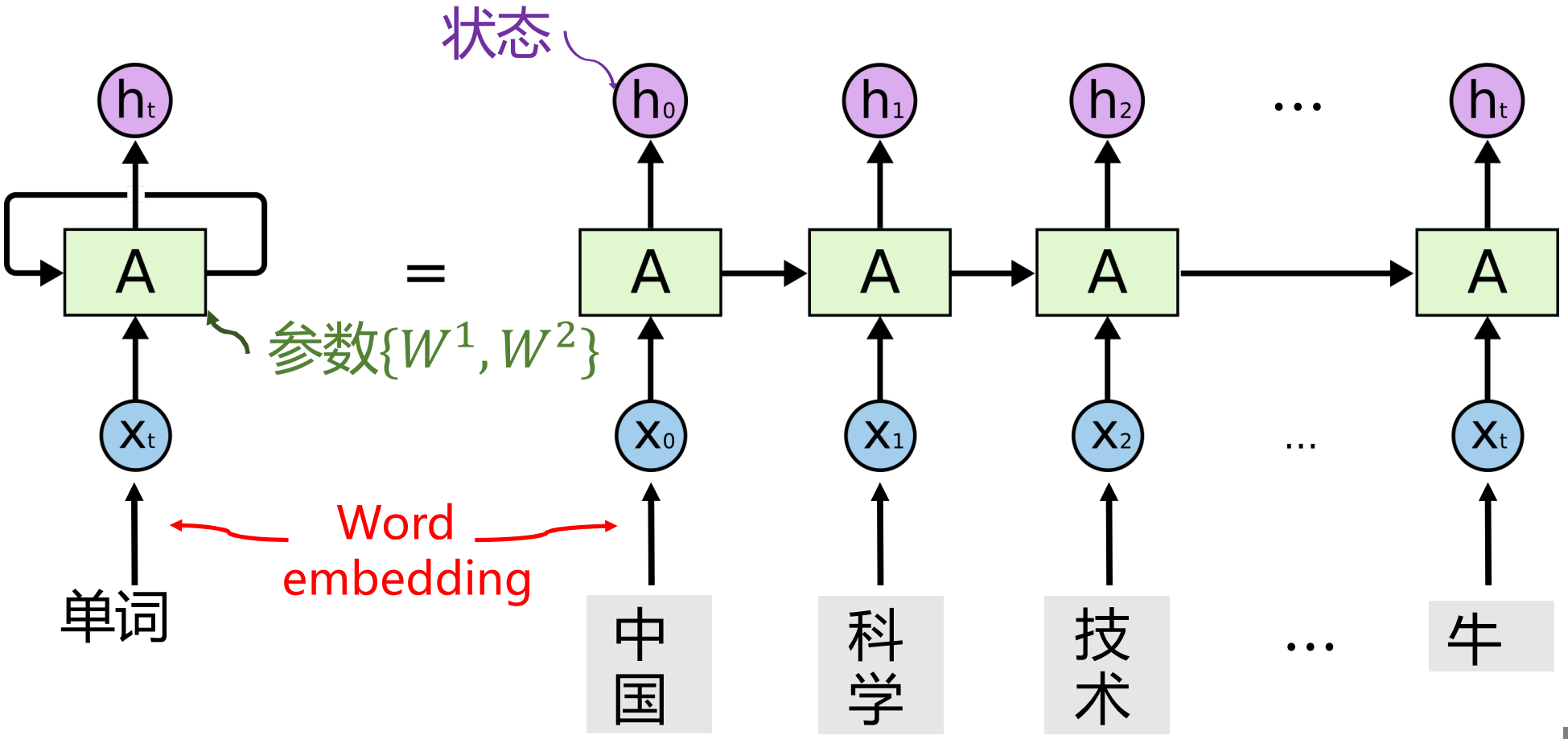
- 将段落当做整体进行处理
- 输入的维度固定（例如：图像）
- 输出的维度固定（例如：Logits）



循环神经网络RNN

- 按照**词序**依次组合获取文本表示：
  - 每个时刻的表示由此时刻文本和上一时刻隐状态计算得到

$$h_t = \tanh(W^1 h_{t-1} + W^2 x_t)$$

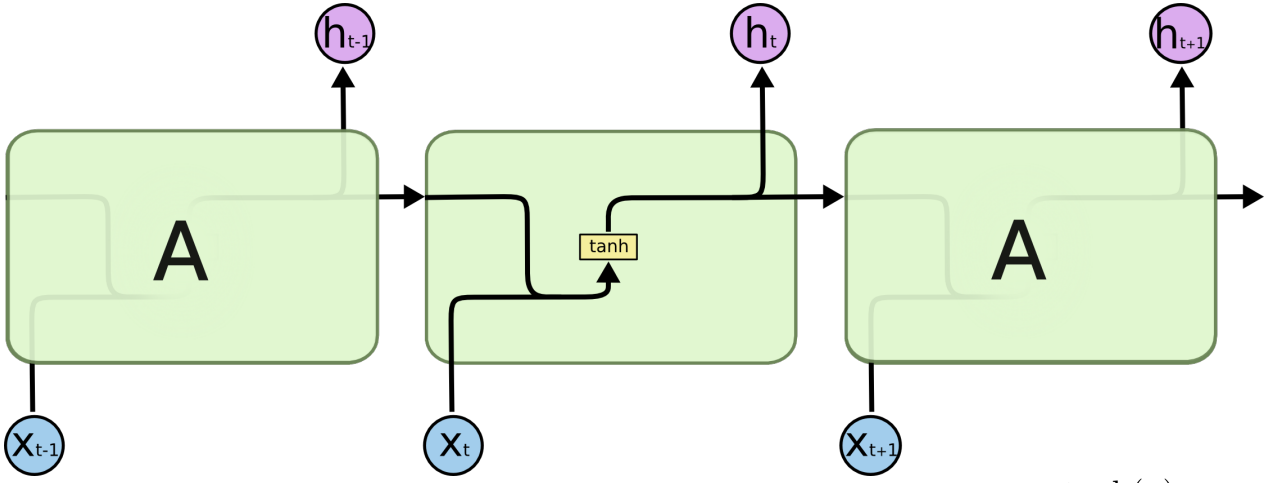


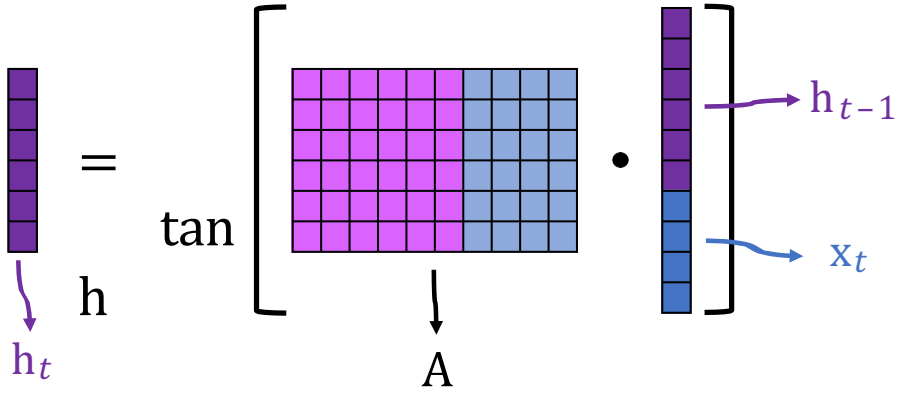
Understanding LSTM Networks.

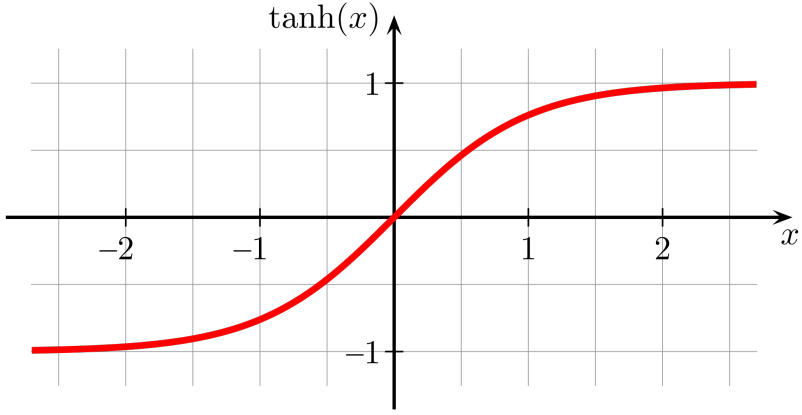
循环神经网络RNN

- 按照**词序**依次组合获取文本表示：
  - 每个时刻的表示由此时刻文本和上一时刻隐状态计算得到

$$h_t = \tanh(W^1 h_{t-1} + W^2 x_t)$$







tanh函数

Understanding LSTM Networks.

循环神经网络RNN

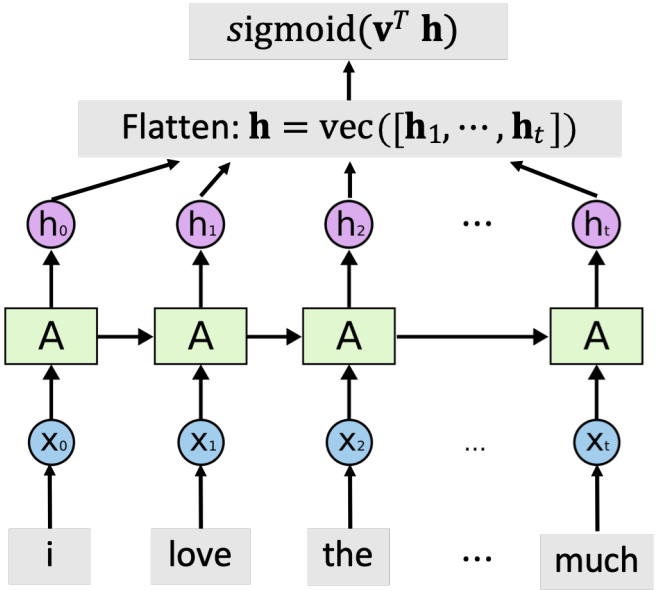
- 按照**词序**依次组合获取文本表示：
  - 每个时刻的表示由此时刻文本和上一时刻隐状态计算得到

$$h_t = \tanh(W^1 h_{t-1} + W^2 x_t)$$

新状态 = f （旧记忆， 新输入）

  - 一套参数，通过时间展开，处理任意长度的序列

  - 采用**最后一个词隐状态**或所有词的隐状态**平均池化**或**最大池化**作为句子表示



RNN应用示例：情感分类

情感分类工作流程

① 输入层

将句子中每个词转换为词向量（Embedding）

↓

② RNN逐词处理（隐藏状态持续更新）

这部 h_1 → 电影 h_2 → 真的 h_3 → 很棒 h_4 → h_n 最终状态

↓

③ 分类层

最后一个隐藏状态 h_n → 全连接层 → Softmax → 情感类别

↓

④ 输出结果

P(正面) = 0.89 | P(负面) = 0.11 →

情感：正面 ✓

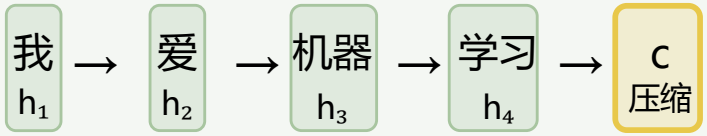
RNN逐个读取词语，**隐藏状态**持续积累句子信息

最后时刻的隐藏状态代表整个句子的语义摘要

分类层将语义向量映射到**情感类别**上

RNN应用示例：机器翻译与Seq2Seq架构

- **编码器 (Encoder)**



输入：源语言句子（中文）

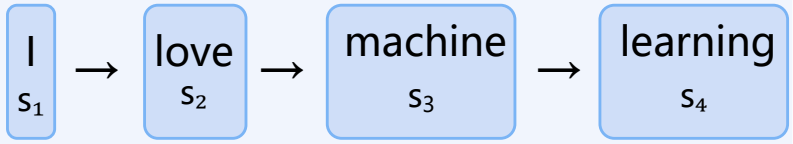
逐词读取源语言，每步更新隐藏状态
最终输出 **上下文向量 c**（整句压缩表示）

上下文向量

c
整句语义压缩向量

信息瓶颈 →

- **解码器 (Decoder)**



输出：目标语言句子（英文）

以 **c** 为初始状态，逐词生成目标语言
每步将已生成词作为下一步输入（自回归）

➢ **Seq2Seq 架构要点**

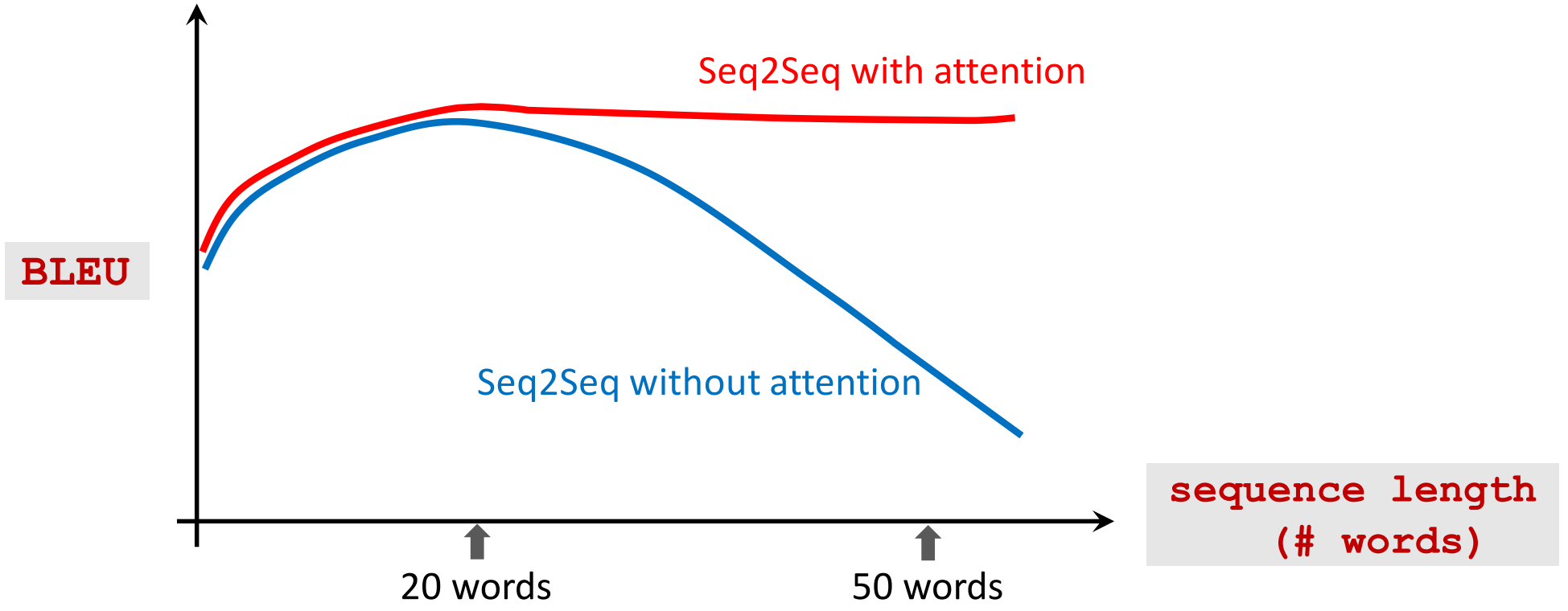
- 编码器与解码器均为RNN（或LSTM）
- 两者通过**上下文向量 c** 连接
- 解码器可以输出与输入不等长的序列

➢ **固定上下文向量的问题**

- 所有信息压缩到**一个固定大小的向量**
- 长句子信息容易丢失，翻译质量下降

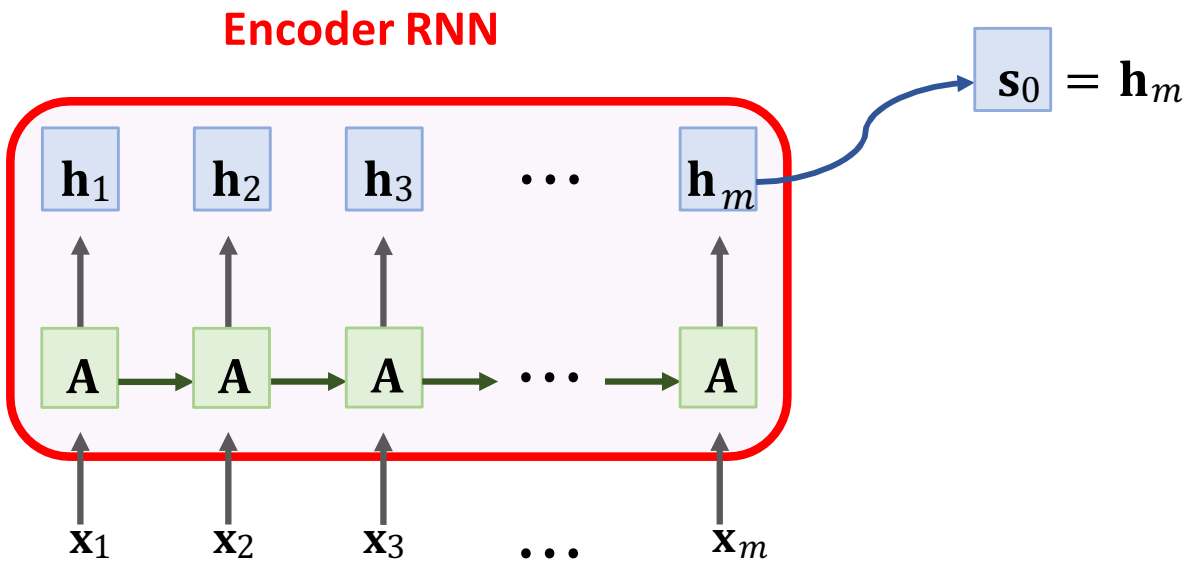
RNN应用示例：机器翻译与Seq2Seq架构

- 长句子信息容易丢失，翻译质量下降
- 利用注意力(attention)机制，对不同词的重要程度加以评估，能够聚焦到更重要的信息



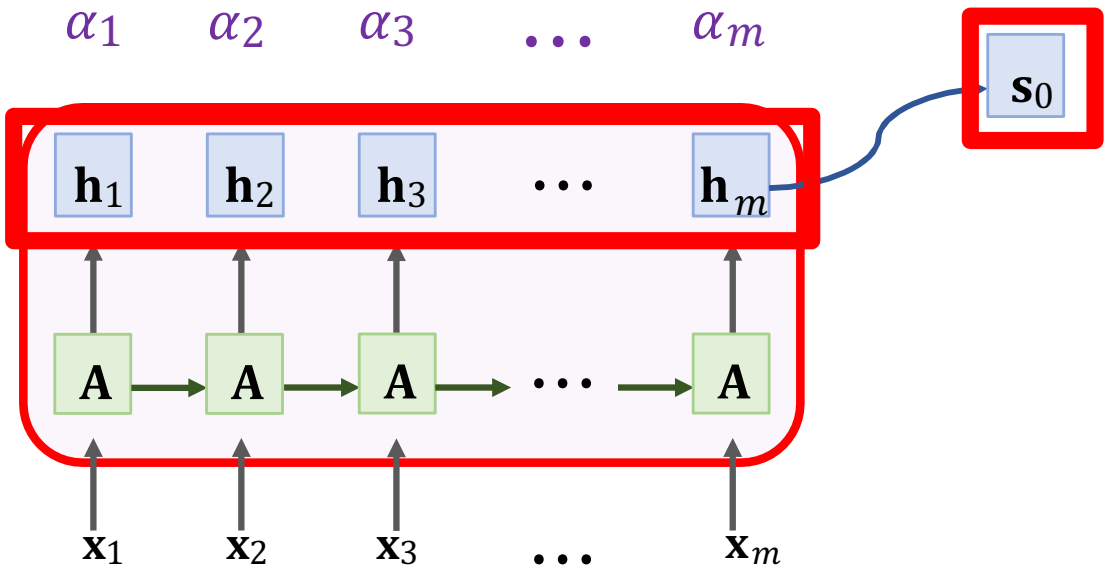
Bahdanau, Cho, & Bengio. Neural machine translation by jointly learning to align and translate. In *ICLR*, 2015.

注意力机制


Encoder RNN
$\mathbf{s}_0 = \mathbf{h}_m$
$\mathbf{h}_1$
$\mathbf{h}_2$
$\mathbf{h}_3$
$\cdots$
$\mathbf{h}_m$
A
A
A
$\cdots$
A
$\mathbf{x}_1$
$\mathbf{x}_2$
$\mathbf{x}_3$
$\cdots$
$\mathbf{x}_m$

注意力机制

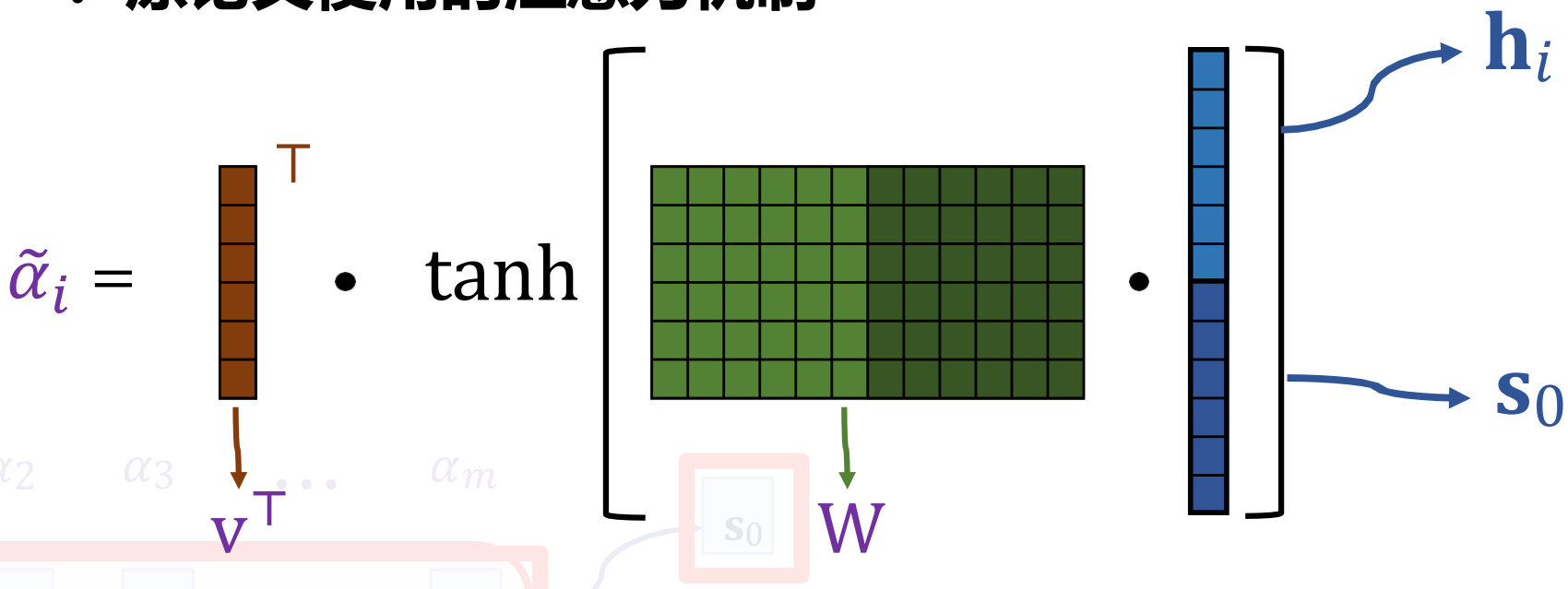
权重：$\alpha_i = \text{align}(\mathbf{h}_i, \mathbf{s}_0)$



注意力机制

权重：$\alpha_i = \text{align}(\mathbf{h}_i, \mathbf{s}_0)$

- **方法一：原论文使用的注意力机制**



然后归一化$\tilde{\alpha}_1, \cdots, \tilde{\alpha}_m$(使得加和为1):

$$\alpha_1, \cdots, \alpha_m = \text{Softmax}(\tilde{\alpha}_1, \cdots, \tilde{\alpha}_m) .$$

注意力机制

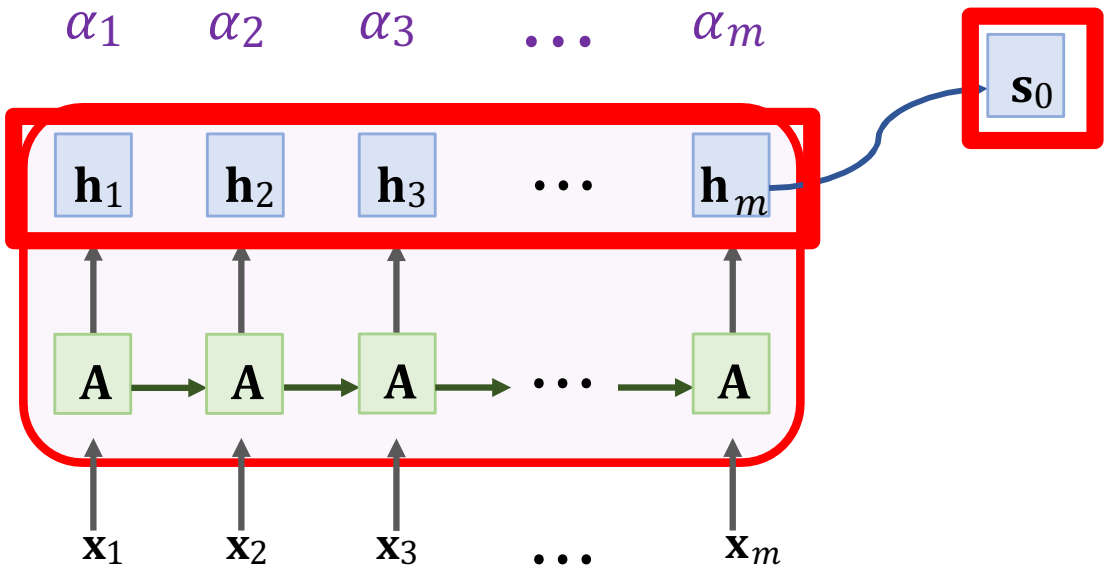
权重：$\alpha_i = \text{align}(\mathbf{h}_i, \mathbf{s}_0)$

- **方法二：与Transformer一致**

1. 线性映射
   - Query: $\mathbf{q}_0 = \mathbf{W}_Q \mathbf{s}_0$. (To match others.)
   - Key: $\mathbf{k}_i = \mathbf{W}_K \mathbf{h}_i$. (To be matched.)
   - Value: $\mathbf{v}_i = \mathbf{W}_V \mathbf{s}_i$. (To beweighted averaged)
2. 内积
   - $\tilde{\alpha}_i = \mathbf{k}_i^\top \mathbf{q}_0$, for $i = 1, \cdots, m$
3. 归一化
   - $[\alpha_1, \cdots, \alpha_m] = \text{Softmax}([\tilde{\alpha}_1, \cdots, \tilde{\alpha}_m])$

注意力机制

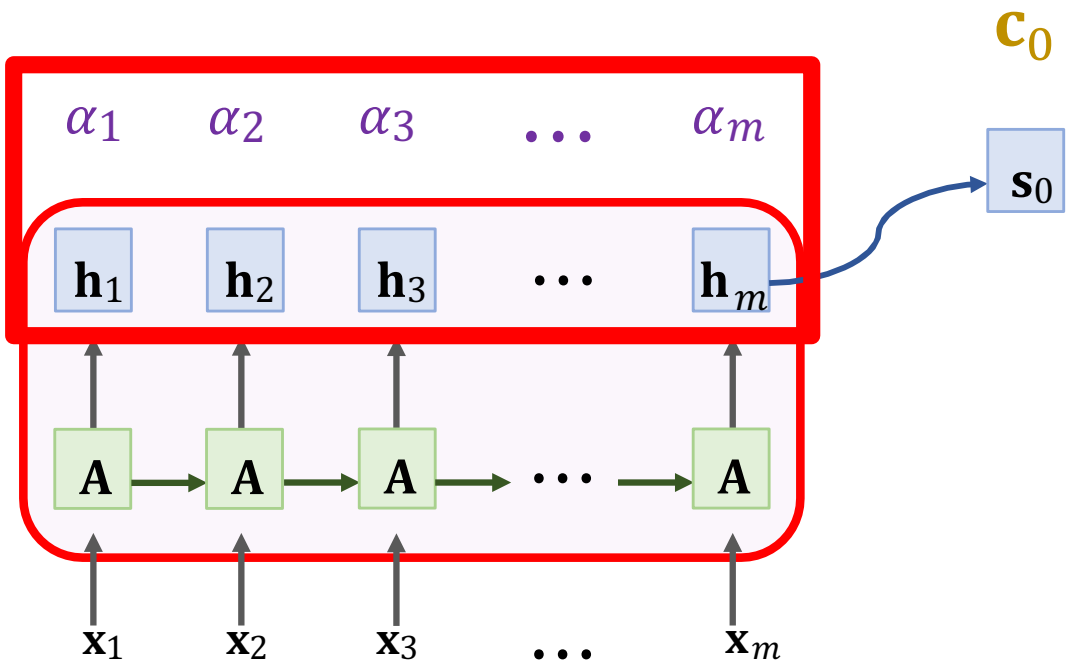
权重：$\alpha_i = \text{align}(\mathbf{h}_i, \mathbf{s}_0)$



注意力机制

权重：$\alpha_i = \text{align}(\mathbf{h}_i, \mathbf{s}_0)$

上下文向量：$\mathbf{c}_0 = \alpha_1 \mathbf{h}_1 + \cdots + \alpha_m \mathbf{h}_m$



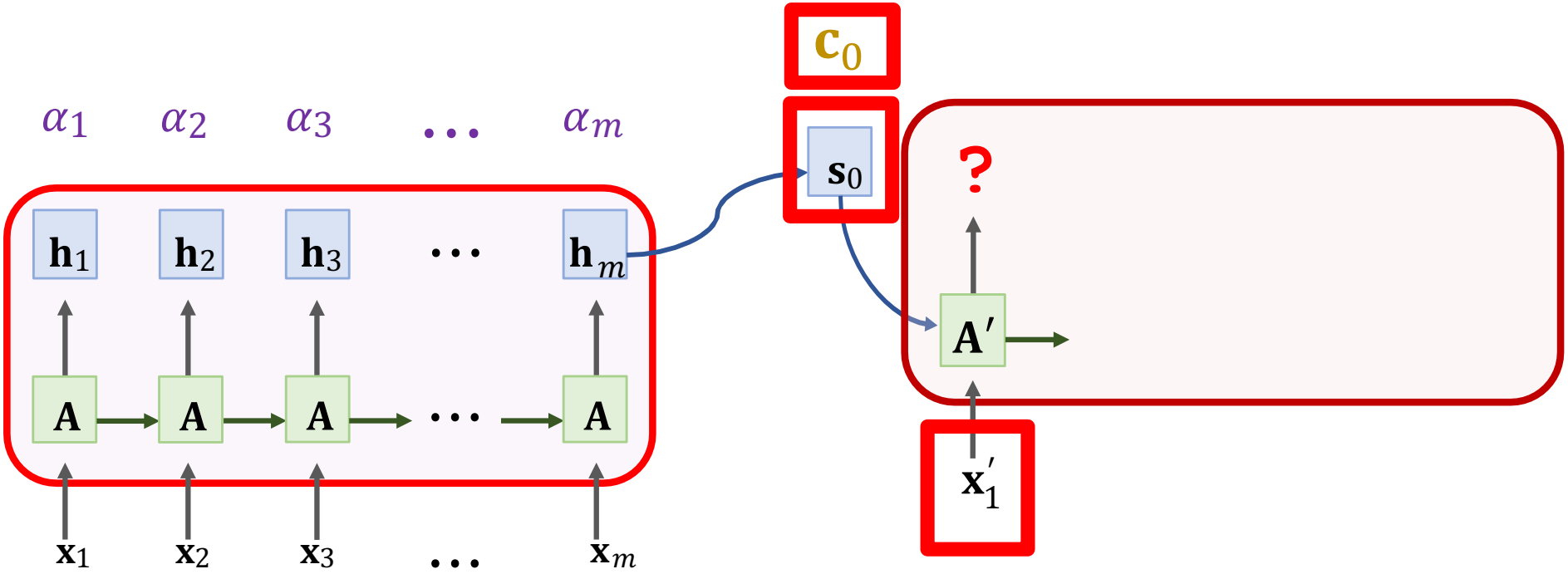
注意力机制

RNN:

$$s_1 = \tanh\left(A' \cdot \begin{bmatrix} x_1' \\ s_0 \end{bmatrix}\right) + b$$

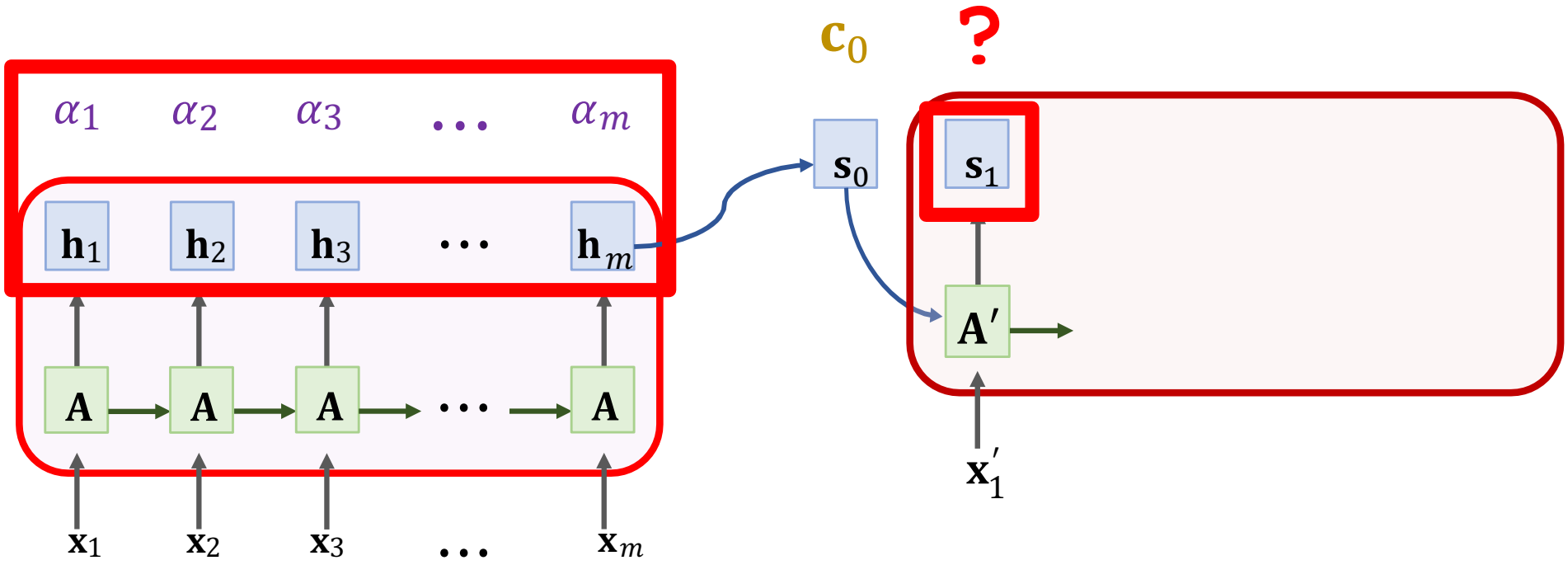
RNN + Attention:

$$s_1 = \tanh\left(A' \cdot \begin{bmatrix} x_1' \\ s_0 \\ c_0 \end{bmatrix}\right) + b$$



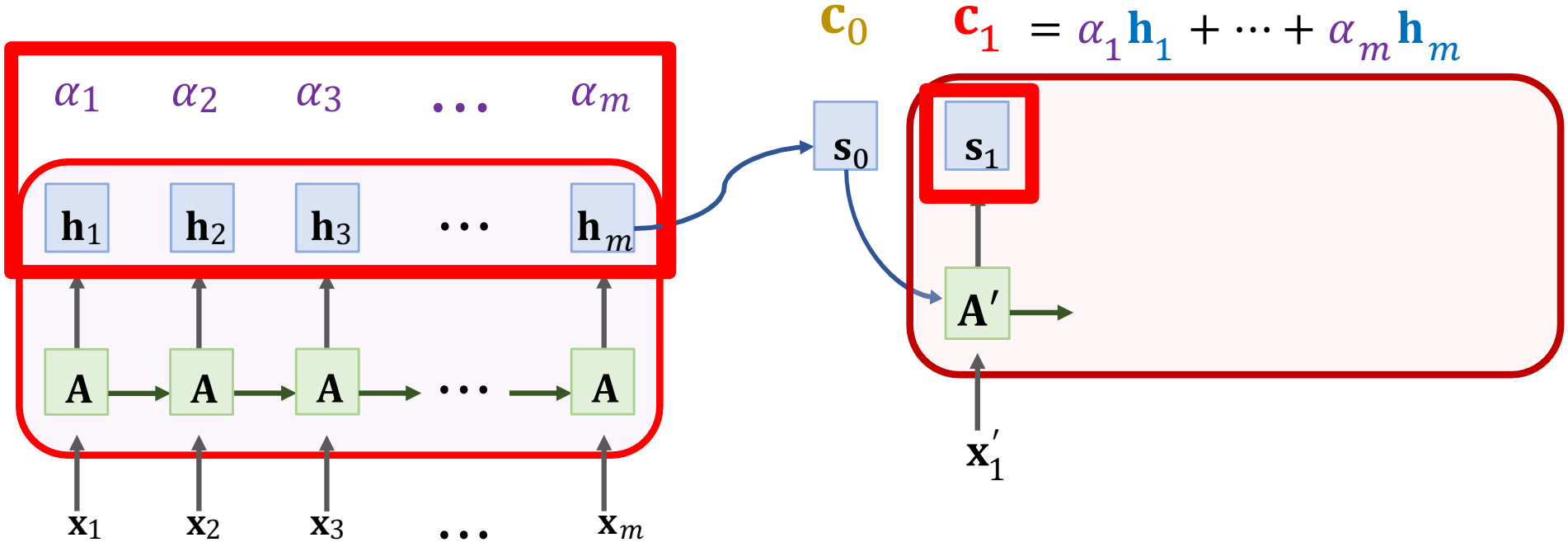
注意力机制

权重：$\alpha_i = \text{align}(\mathbf{h}_i, \mathbf{s}_1)$



注意力机制

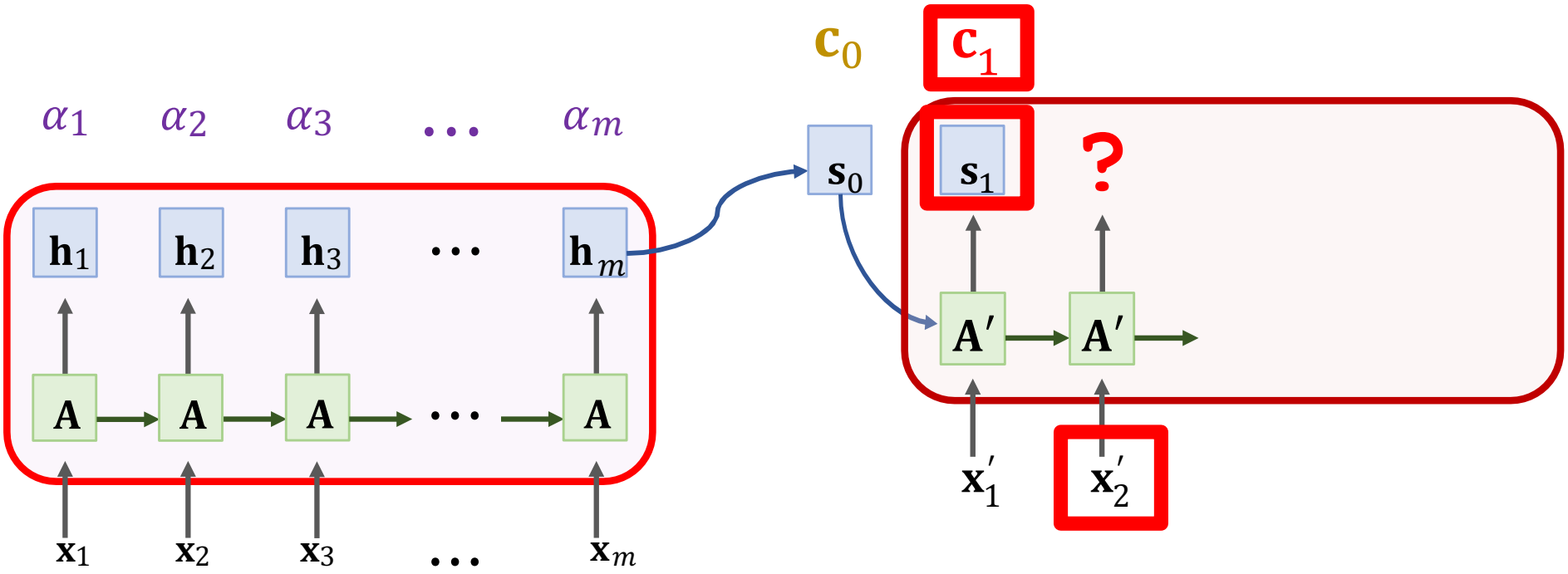
$$\text{权重：}\ \alpha_i = \text{align}(\mathbf{h}_i, \mathbf{s}_1)$$



注意力机制

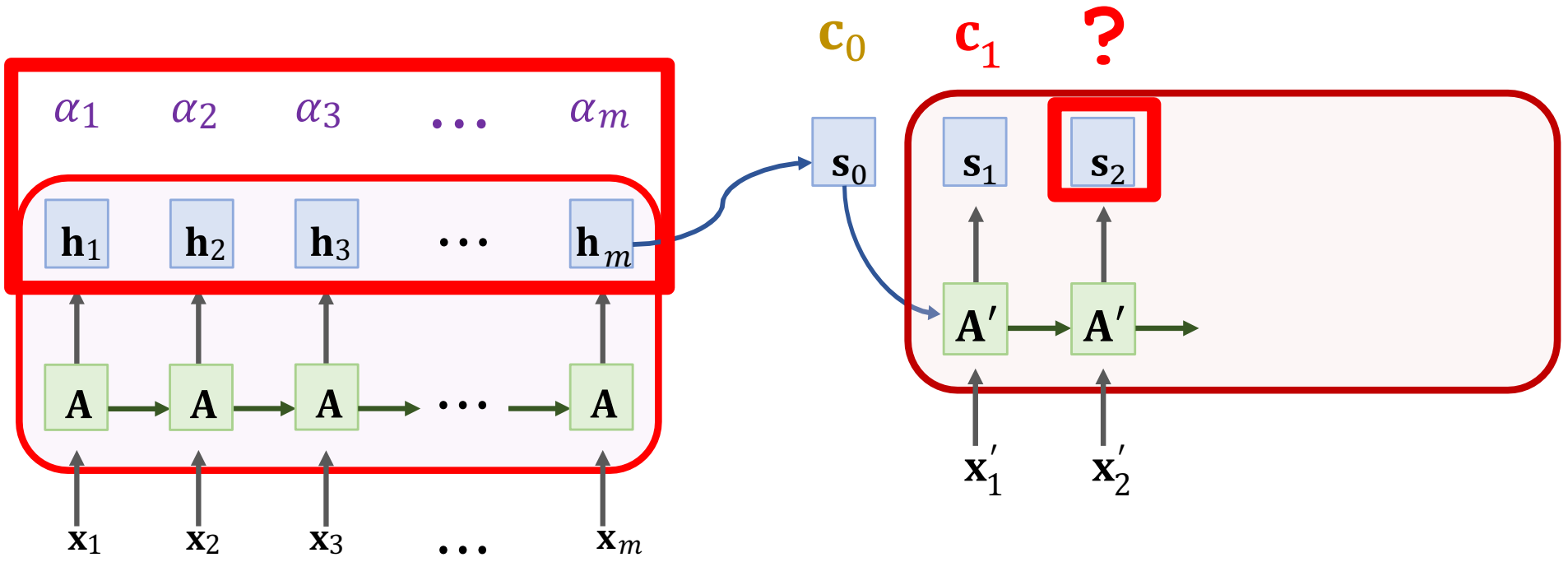
RNN + Attention:

$$\mathbf{s}_2 = \tanh\left(A' \cdot \begin{bmatrix} x'_2 \\ \mathbf{s}_1 \\ \mathbf{c}_1 \end{bmatrix}\right) + \mathbf{b}$$



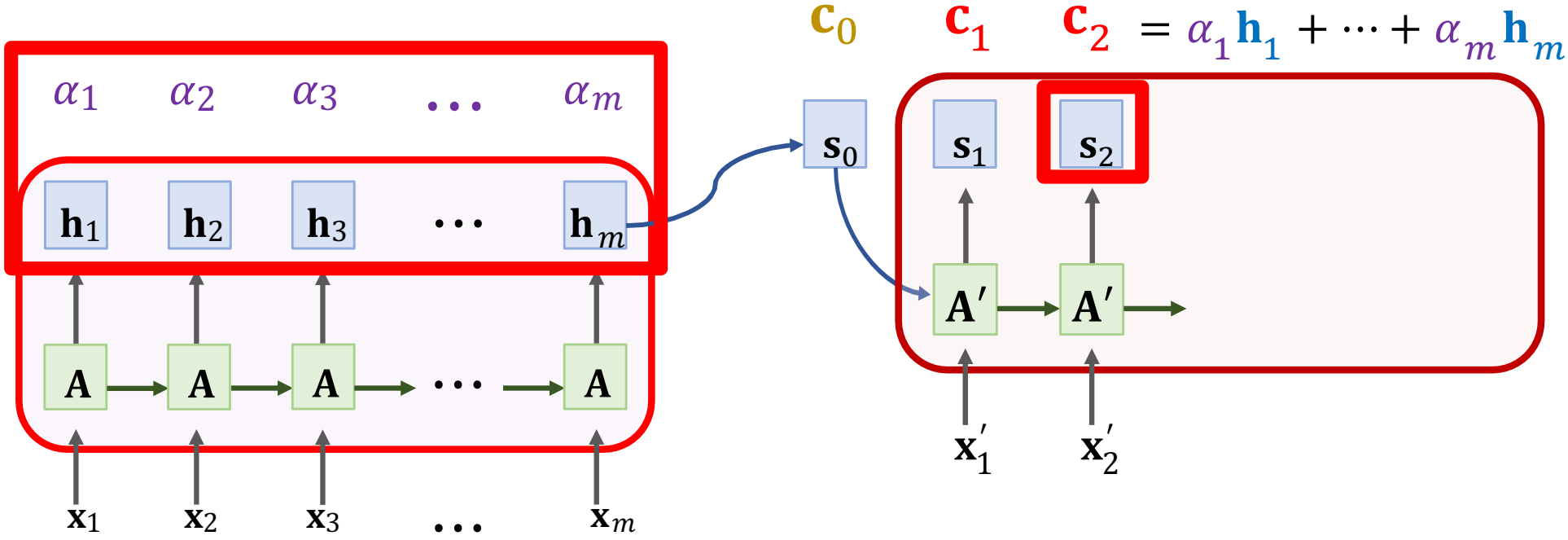
注意力机制

$$\text{权重：}\ \alpha_i = \text{align}(\mathbf{h}_i, \mathbf{s}_2)$$



注意力机制

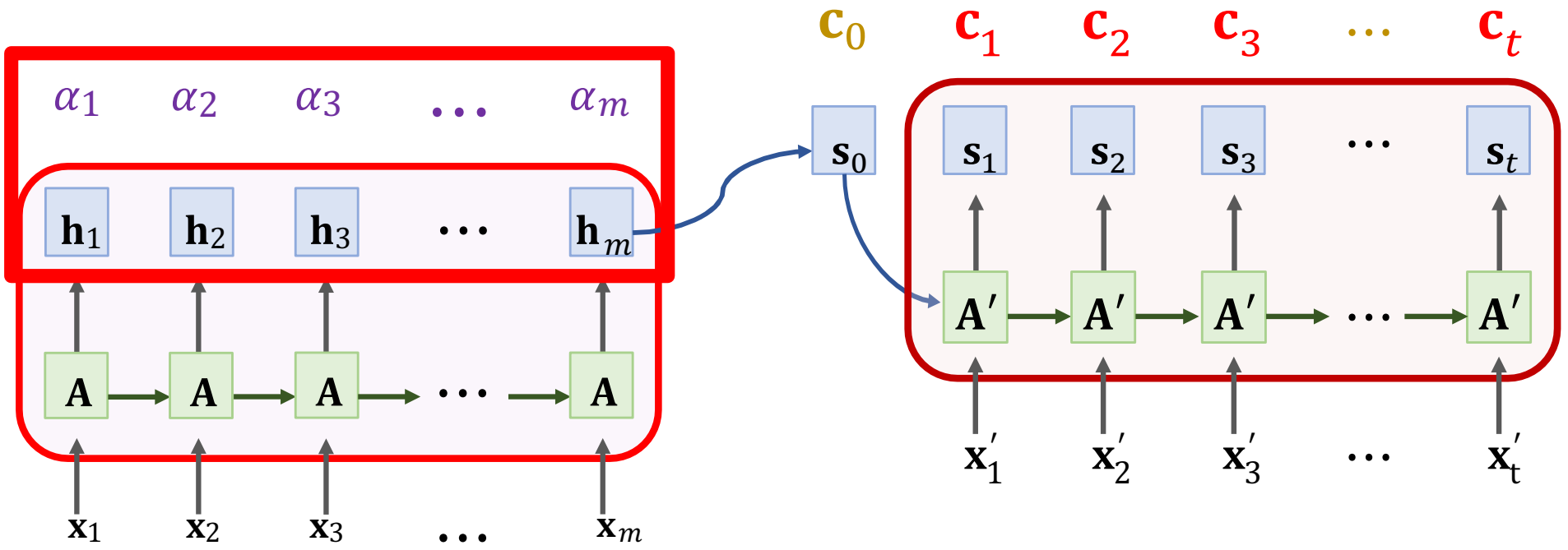
$$\text{权重：}\ \alpha_i = \text{align}(\mathbf{h}_i, \mathbf{s}_2)$$



注意力机制

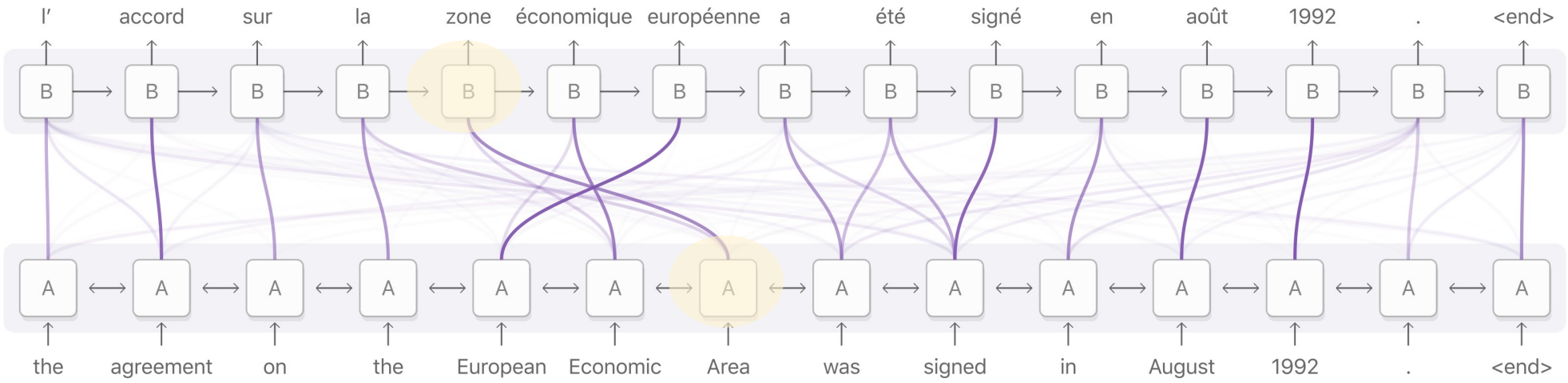
问题：权重α_i被计算了多少次?

- 每计算一个$\mathbf{c}_j$，需要计算m个权重$\alpha_1, \cdots, \alpha_m$.
- 解码器共t个状态，因此需要计算mt个权重.



注意力机制

Decoder RNN （目标语言：法语）



Encoder RNN （源语言：英语）

https://distill.pub/2016/augmented-rnns/

RNN的梯度消失/梯度爆炸问题

- 训练时需要用**反向传播**计算损失 $J(\theta)$ 对早期参数的梯度，经过链式法则逐步回传：

$$\frac{\partial J}{\partial h_1} = \frac{\partial J}{\partial h_T} \cdot \frac{\partial h_T}{\partial h_{T-1}} \cdot \frac{\partial h_{T-1}}{\partial h_{T-2}} \cdots \frac{\partial h_2}{\partial h_1}$$

- 每一步的局部梯度为：

$$\frac{\partial h_t}{\partial h_{t-1}} = W_h^T \cdot \mathrm{diag}\left(\tanh'(z_t)\right)$$

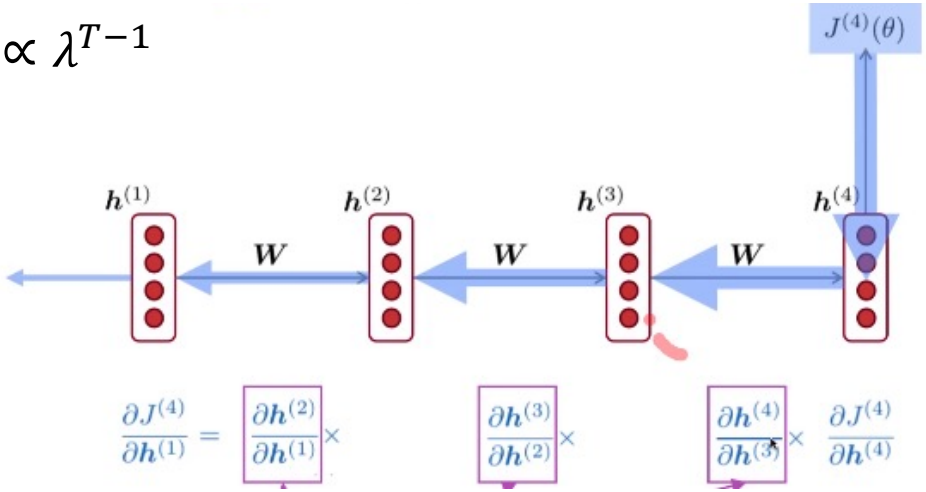
- 经过 T 步连乘后

$$\frac{\partial L}{\partial h_1} \propto (W_h \cdot \tanh')^{T-1}$$

- 设 λ 为矩阵 W_h 的特征值量级

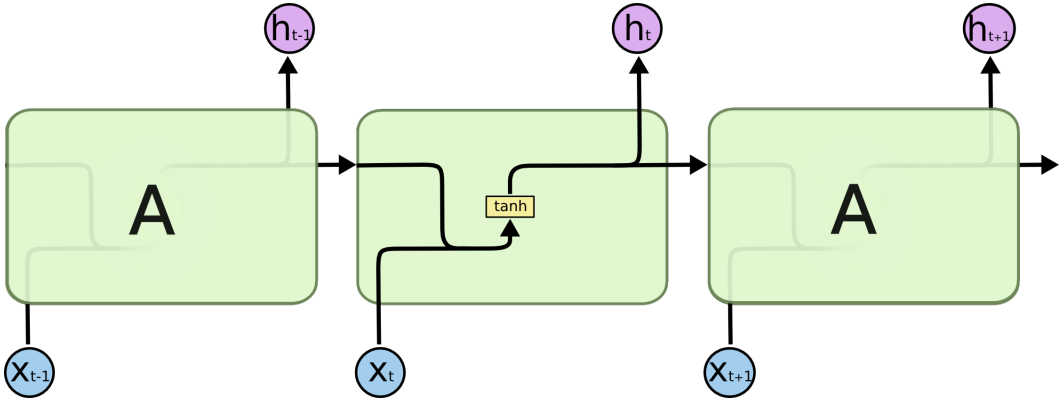
$$\text{梯度} \propto \lambda^{T-1}$$

情形	结果	问题
$\lambda < 1$	$\lambda^{T-1} \to 0$	**梯度消失**，早期层几乎学不到东西
$\lambda > 1$	$\lambda^{T-1} \to \infty$	**梯度爆炸**，训练发散

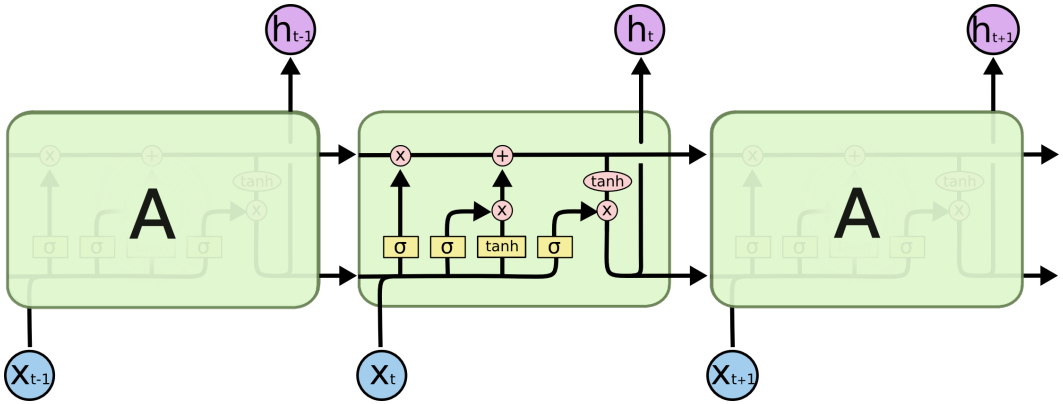


LSTM：长短期记忆网络与门控机制

- RNN 用单一隐状态承载所有记忆，难以处理长程依赖关系，且无法区分“该记什么”与“该忘什么”
- LSTM引入三种门控机制， 精确控制信息的**记忆、遗忘和输出**



RNN

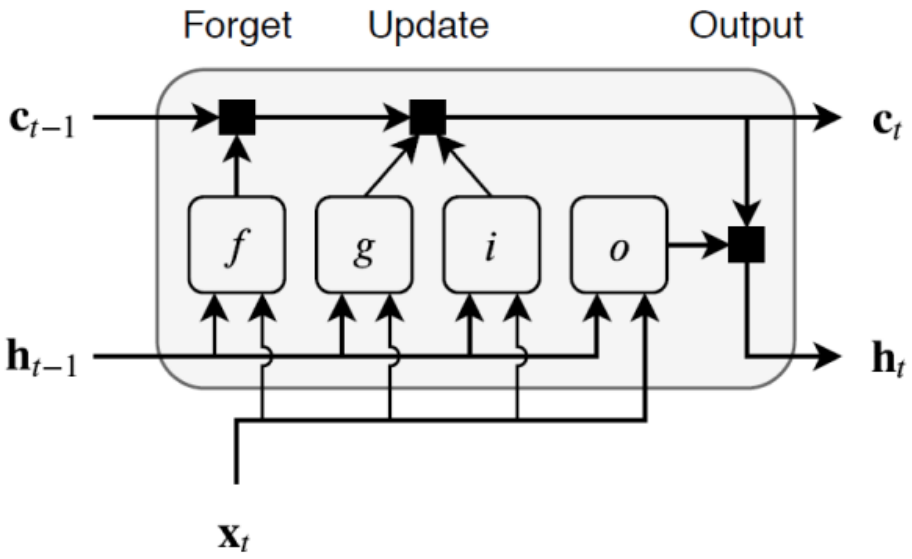


LSTM

图片摘自博客: Understanding LSTM Networks.

LSTM：长短期记忆网络与门控机制

➢ LSTM单步工作流程



1. **遗忘**：看看旧记忆里哪些已经没用了，用遗忘门打分决定保留比例
2. **更新**：根据当前输入，准备新信息，通过输入门写入细胞状态
3. **输出**：从更新后的细胞状态中，通过输出门筛选需要传递的信息

➢ 类比：工厂传送带 + 三个工人

旧记忆 C_{t-1}

🗑️ **工人①**
遗忘门

✏️ **工人②**
输入门

📤 **工人③**
输出门

h_t
输出

细胞状态传送带（长期记忆）

📦 旧货物

扔掉
过期货物

放入
新货物

取出需要
的货物

LSTM：长短期记忆网络与门控机制

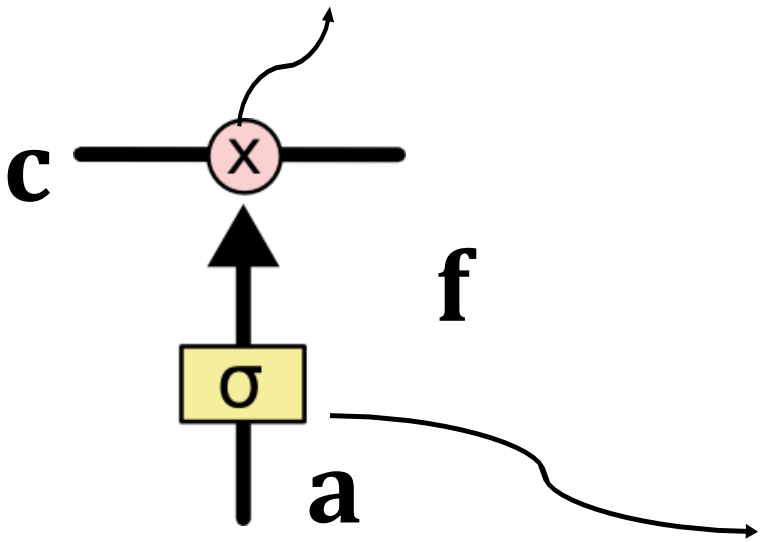
- **遗忘门 (Forget Gate) —— 决定丢弃哪些旧信息**

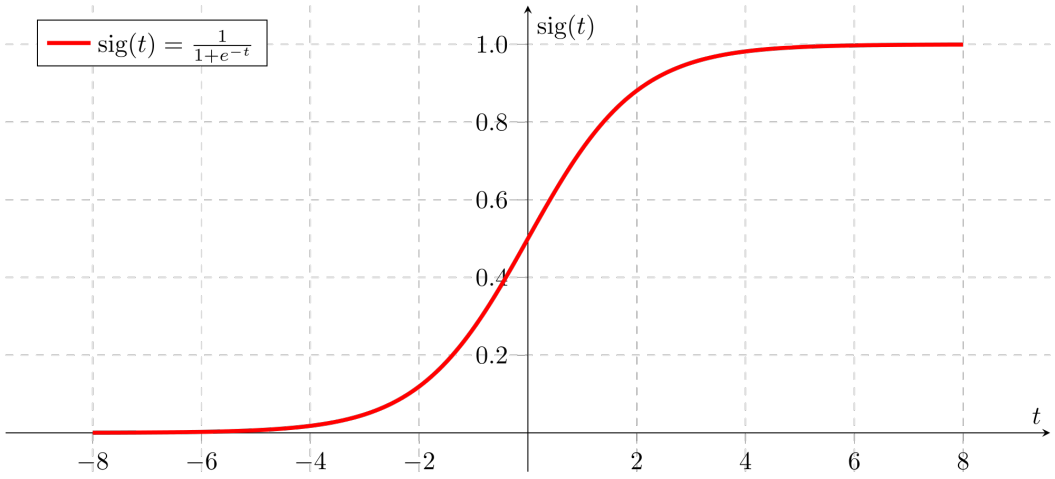
  读取上一时刻隐藏状态 h_{t-1} 和当前输入 x_t，输出 **0~1 之间的值**：

  0代表“完全遗忘”，1代表“完全保留”

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$

逐元素乘法





Sigmoid函数：值域(0,1)

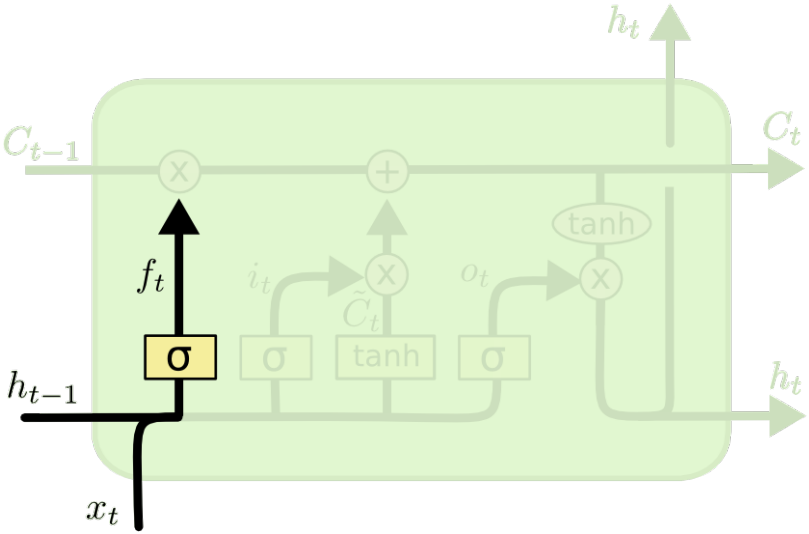
LSTM：长短期记忆网络与门控机制

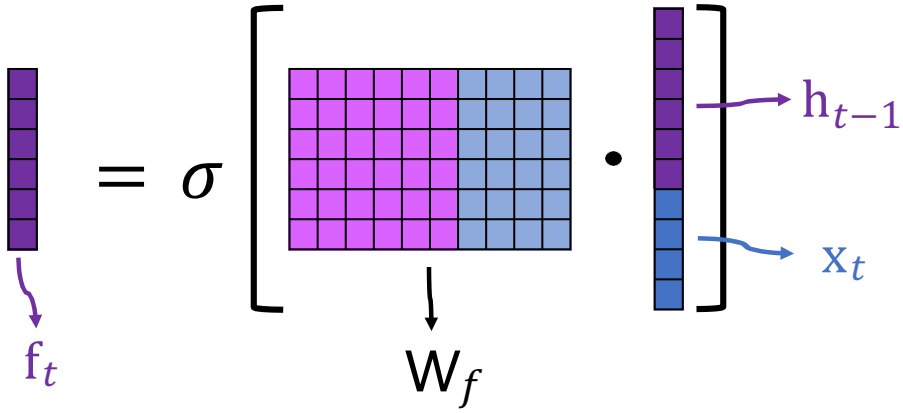
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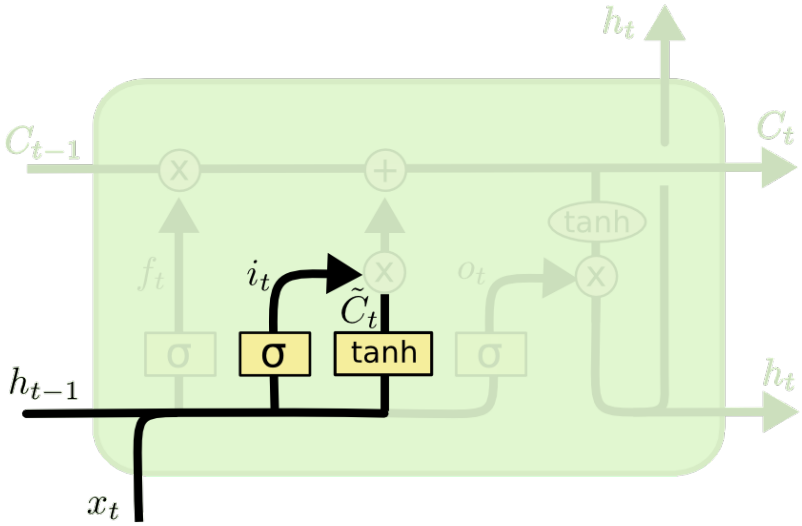
LSTM：长短期记忆网络与门控机制

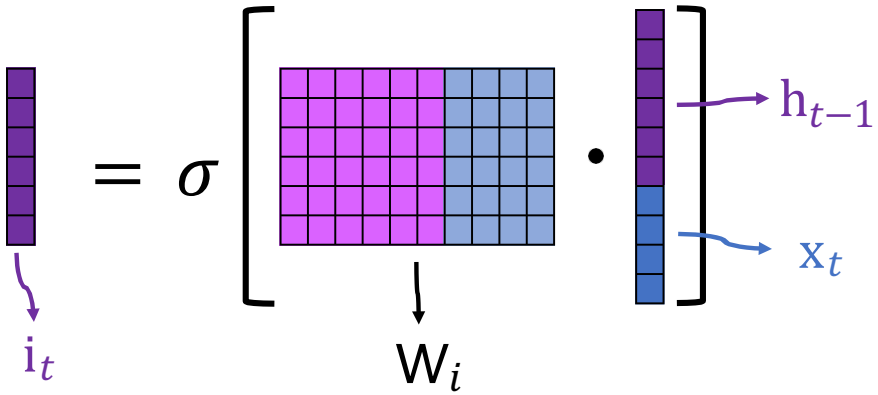
- **输入门 (Input Gate) —— 决定记住哪些新信息**

  两步操作：① 决定**哪些位置需要更新**；② 创建候选新记忆 $\tilde{C}$。

  两者结合写入细胞状态

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$





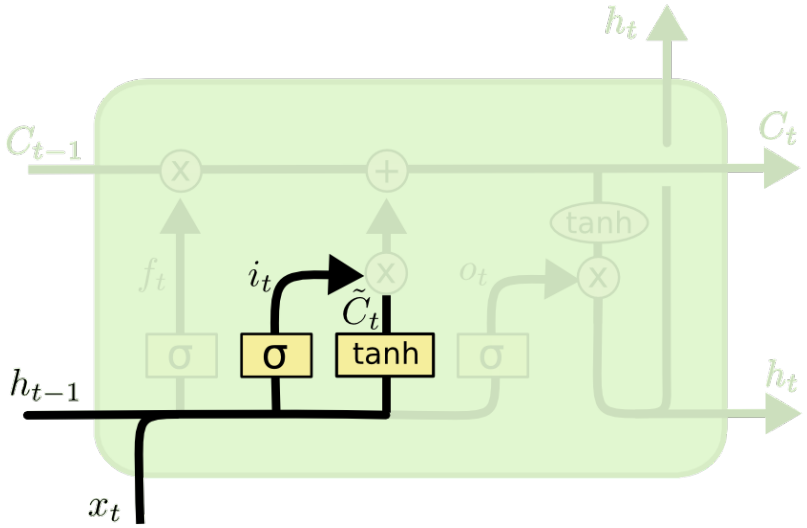
LSTM：长短期记忆网络与门控机制

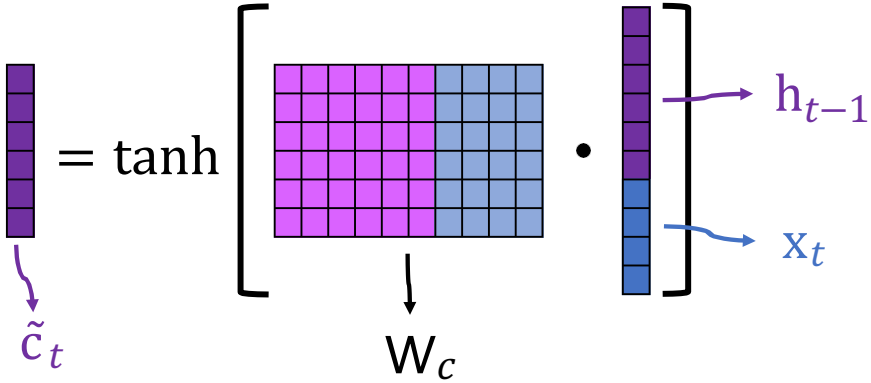
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两者结合写入细胞状态

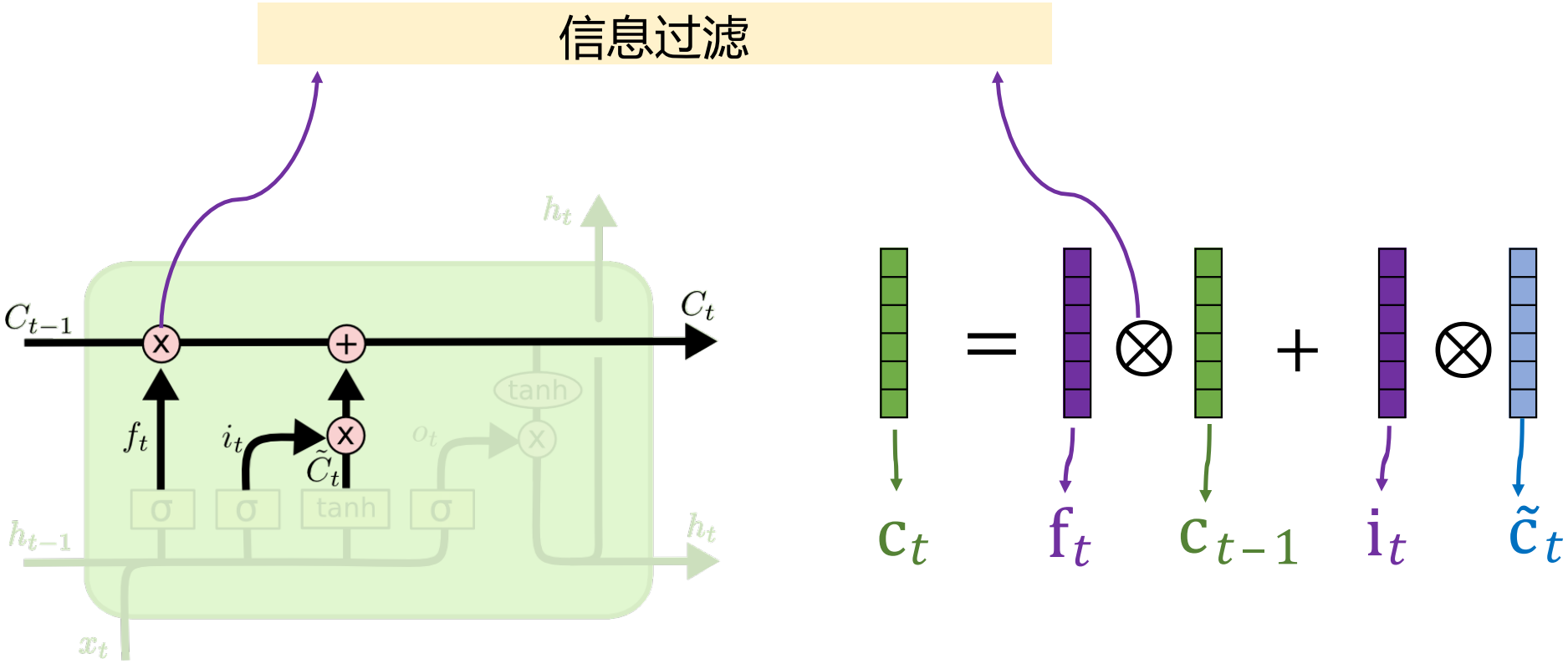
$$\tilde{C}_t = \tanh(W_i \cdot [h_{t-1}, x_t] + b_i)$$





LSTM：长短期记忆网络与门控机制

- 更新Cell状态C_t

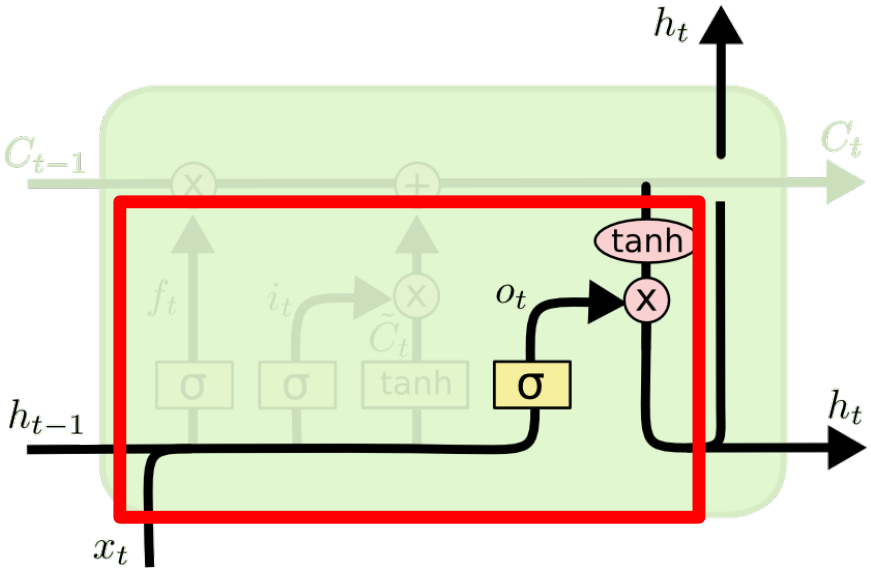


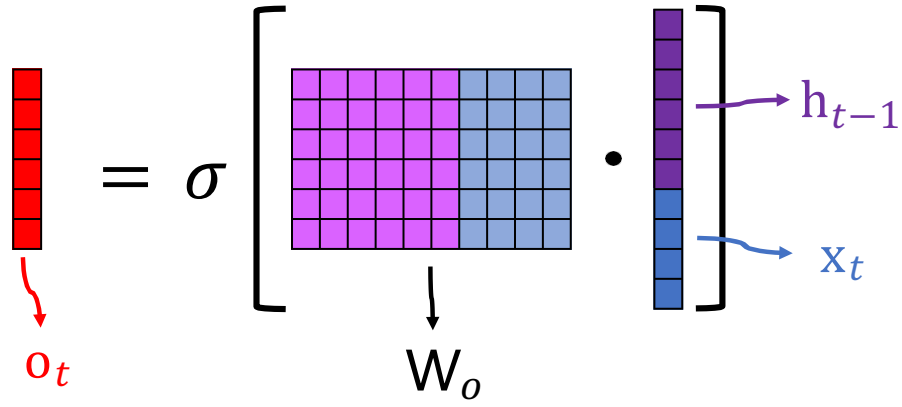
LSTM：长短期记忆网络与门控机制

➢ **输出门 (Output Gate)** —— 决定输出细胞状态的哪些部分

基于当前输入和旧隐藏状态，决定**哪些细胞状态值**会输出为新的隐藏状态 h_t

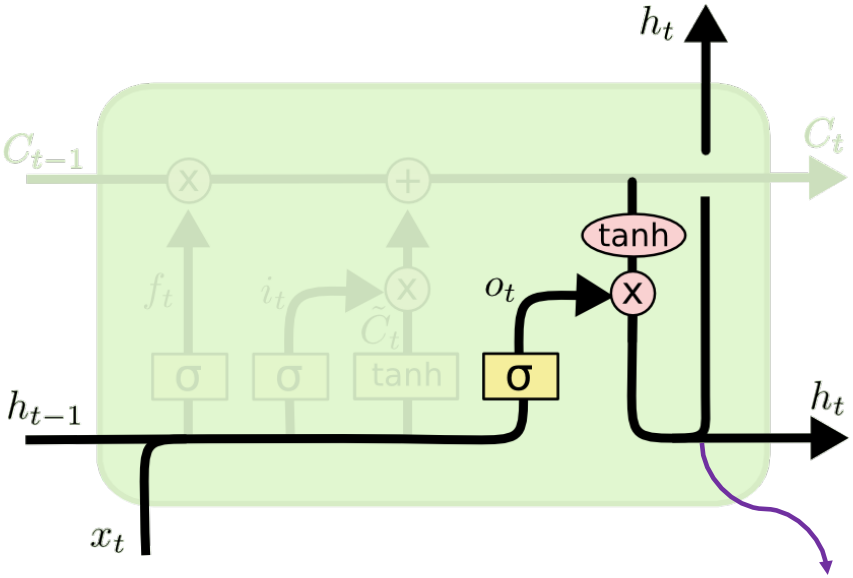
$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$$





LSTM：长短期记忆网络与门控机制

- 更新隐层状态h_t



两份h_t

$$\mathbf{h}_t = \mathbf{o}_t \otimes \tanh(\mathbf{c}_t)$$

LSTM：长短期记忆网络与门控机制

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$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$$

LSTM的局限性

- **仍然难以捕捉极长距离依赖**
  - LSTM 虽然缓解了梯度消失，但本质上还是**顺序传递信息**，序列越长，早期信息经过多次门控衰减后仍然会逐渐淡化
  - 对于段落级、文档级的长文本，效果依然有限
- **无法并行计算**
  - LSTM 必须**从左到右逐步计算**，h_t 依赖 h_{t-1}，天然串行。这意味着：
    - 无法利用 GPU 的并行能力
    - 训练速度慢，难以扩展到大规模数据
- **参数复杂，调参困难**
  - LSTM 有输入门、遗忘门、输出门、Cell状态四套参数，结构复杂，训练时更容易出现不稳定，调参成本高

提纲

1 语言表示与词向量

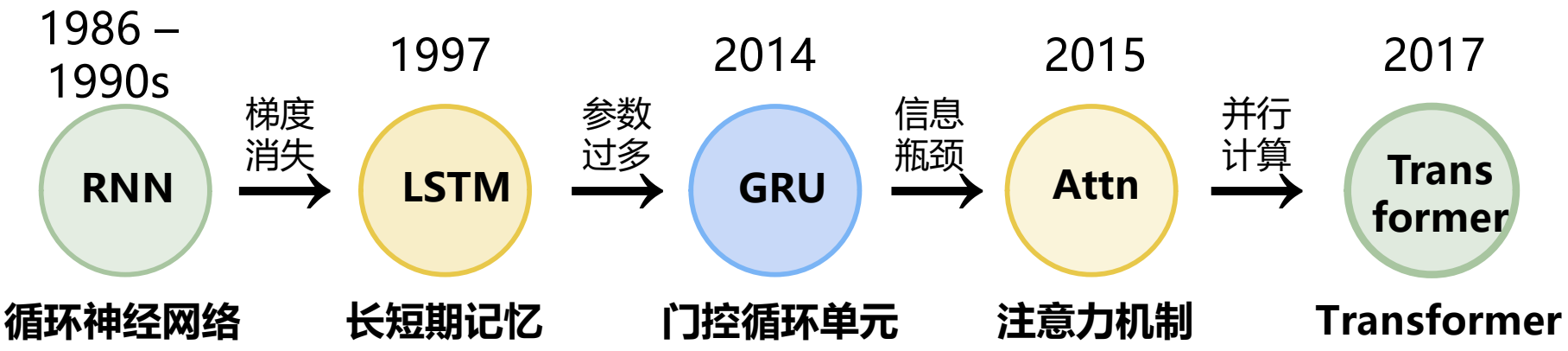
2 序列模型与注意力机制

3 **Transformer与预训练模型**

4 上下文工程

5 大语言模型后训练

序列模型：从RNN到Transformer



🔵 RNN

✅ **优势**
结构简单，参数少；可以处理变长序列

❌ **局限**
梯度消失；无法并行；长程依赖能力弱

核心思想：**隐藏状态记忆**

🟡 LSTM / GRU

✅ **优势**
门控机制解决梯度消失；长程记忆大幅改善

❌ **局限**
仍是串行；计算慢；极长序列仍有局限

核心思想：**门控记忆写入**

🔵 Attention

✅ **优势**
全局关注；可解释；配合RNN使用

❌ **局限**
早期仍依赖RNN；复杂度O(n^2)（后有改进）

核心思想：**动态全局加权**

⚡ Transformer

✅ **优势**
完全并行；多头注意力；可扩展到超大规模

⚠️ **代价**
计算复杂度O(n^2)；需要海量数据；参数多

核心思想：**纯注意力架构**

Transformer的诞生

"Attention is All You Need"

In NIPS, 2017 · Google Brain / Google Research

Author List: Vaswani, Shazeer, Parmar, Uszkoreit, Jones, Gomez, Kaiser, Polosukhin

➢ Transformer的四大创新

- **完全抛弃RNN**，仅用注意力机制建模序列，不再依赖序列循环，解放并行能力
- **多头注意力**，从多个子空间并行捕捉不同关系，如同用多个视角同时审视一句话
- **位置编码**，显式注入顺序信息，利用正弦/余弦函数编码相对位置关系
- **残差连接 + LayerNorm**，稳定深层网络训练，使得堆叠多层成为可能，规模可扩展

➢Transformer催生的技术演进

2017 Transformer原始论文，用于机器翻译

2018 BERT（Google）、GPT-1（OpenAI）相继发布

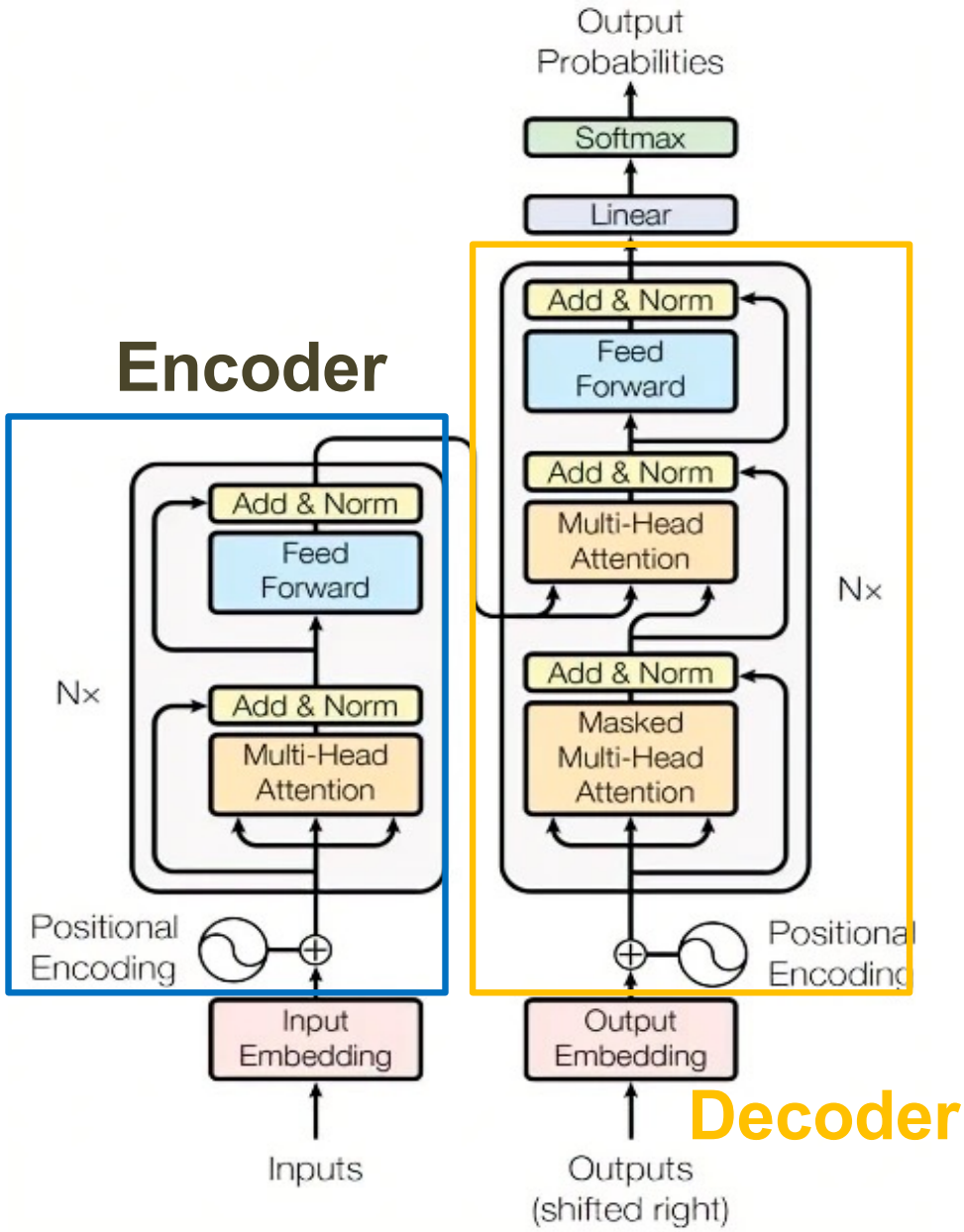
2019 GPT-2、T5、RoBERTa等大量衍生模型

2020 GPT-3（175B参数），in-context learning

2022+ ChatGPT、GPT-4、Gemini、Claude等大模型时代

Transformer结构

- Transformer总体上遵循encoder和decoder的架构；
- Transformer中的encoder和decoder是由以attention为核心的block堆叠而成。

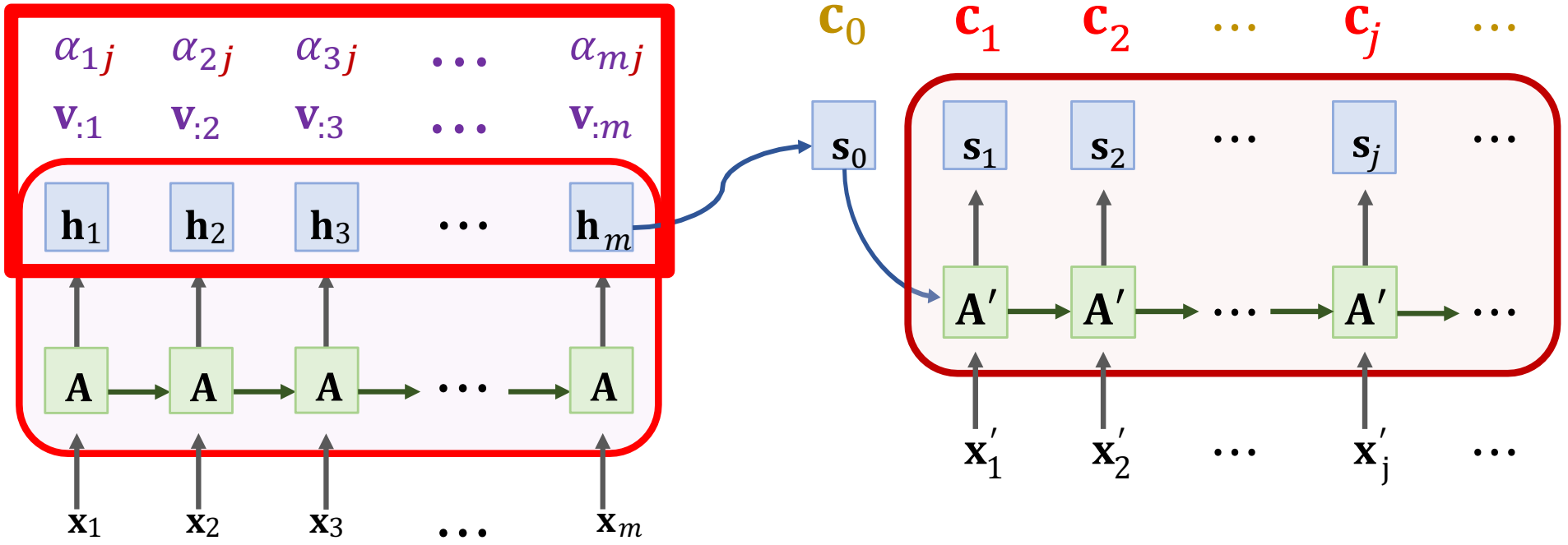


Attention w/ RNN

Query: $\mathbf{q}_{:j} = \mathbf{W}_Q \mathbf{s}_j$, Key: $\mathbf{k}_{:i} = \mathbf{W}_K \mathbf{h}_i$, Value: $\mathbf{v}_{:i} = \mathbf{W}_V \mathbf{h}_i$.

权重： $\boldsymbol{\alpha}_{:j} = \text{Softmax}(\mathbf{K}^\top \mathbf{q}_{:j}) \in \mathbb{R}^m$.

上下文向量： $\mathbf{c}_j = \alpha_{1j}\mathbf{v}_{:1} + \cdots + \alpha_{mj}\mathbf{v}_{:m}$.



Attention w/o RNN

- 对于Seq2Seq模型（Encoder+Decoder）
  - Encoder的输入是向量$\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_m$
  - Decoder的输入是向量$\mathbf{x}'_1, \mathbf{x}'_2, \cdots, \mathbf{x}'_t$
  - $\mathbf{C} = \mathrm{Attn}(\mathbf{X}, \mathbf{X}')$，$\Theta = \{\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V\}$

Encoder's inputs:

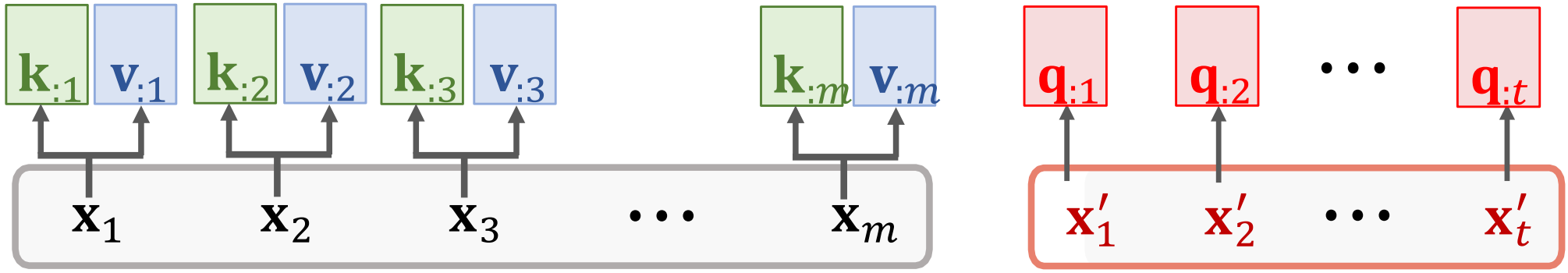
$\mathbf{x}_1$ $\mathbf{x}_2$ $\mathbf{x}_3$ $\cdots$ $\mathbf{x}_m$

Decoder's inputs:

$\mathbf{x}'_1$ $\mathbf{x}'_2$ $\cdots$ $\mathbf{x}'_t$

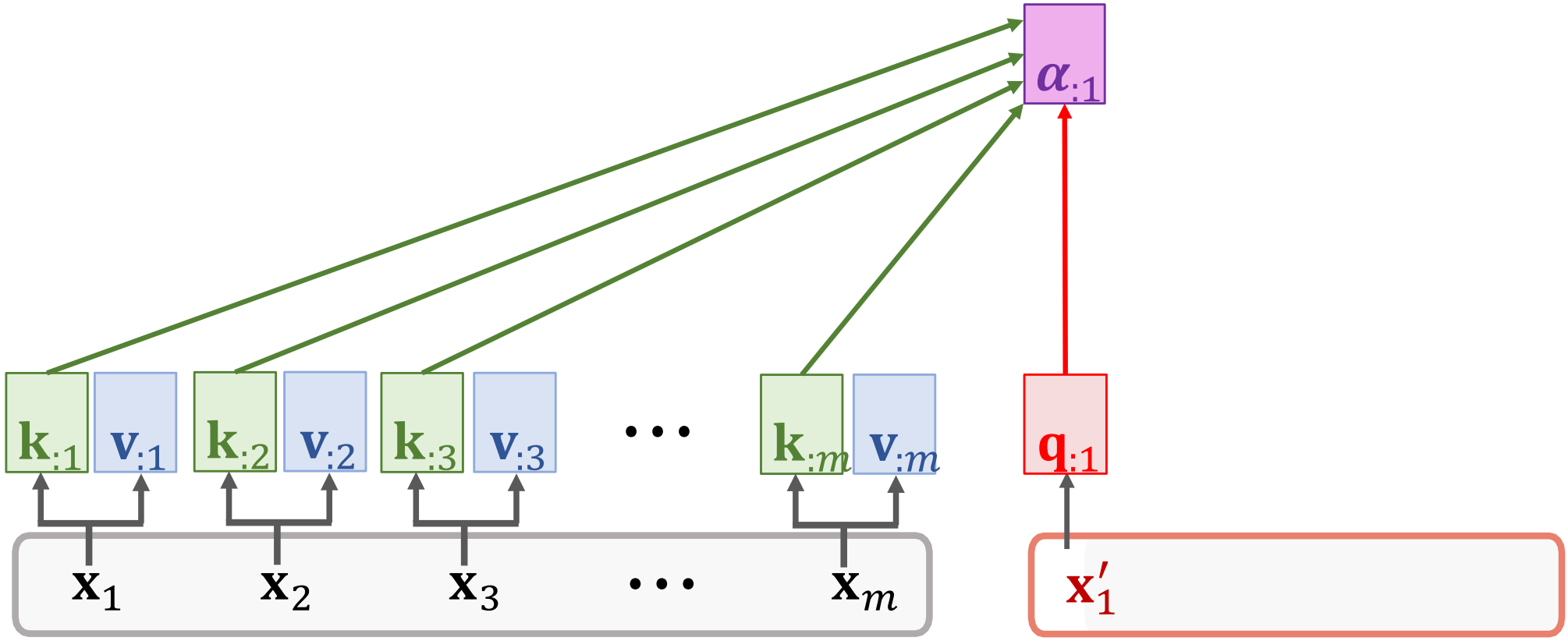
Attention w/o RNN

- Keys和values基于Encoder的输入$\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_m$
  - Key: $\mathbf{k}_{:i} = \mathbf{W}_K \mathbf{x}_i$
  - Value: $\mathbf{v}_{:i} = \mathbf{W}_V \mathbf{x}_i$
- Queries基于Decoder的输入$\mathbf{x}'_1, \mathbf{x}'_2, \cdots, \mathbf{x}'_t$
  - Query: $\mathbf{q}_{:j} = \mathbf{W}_Q \mathbf{x}'_j$



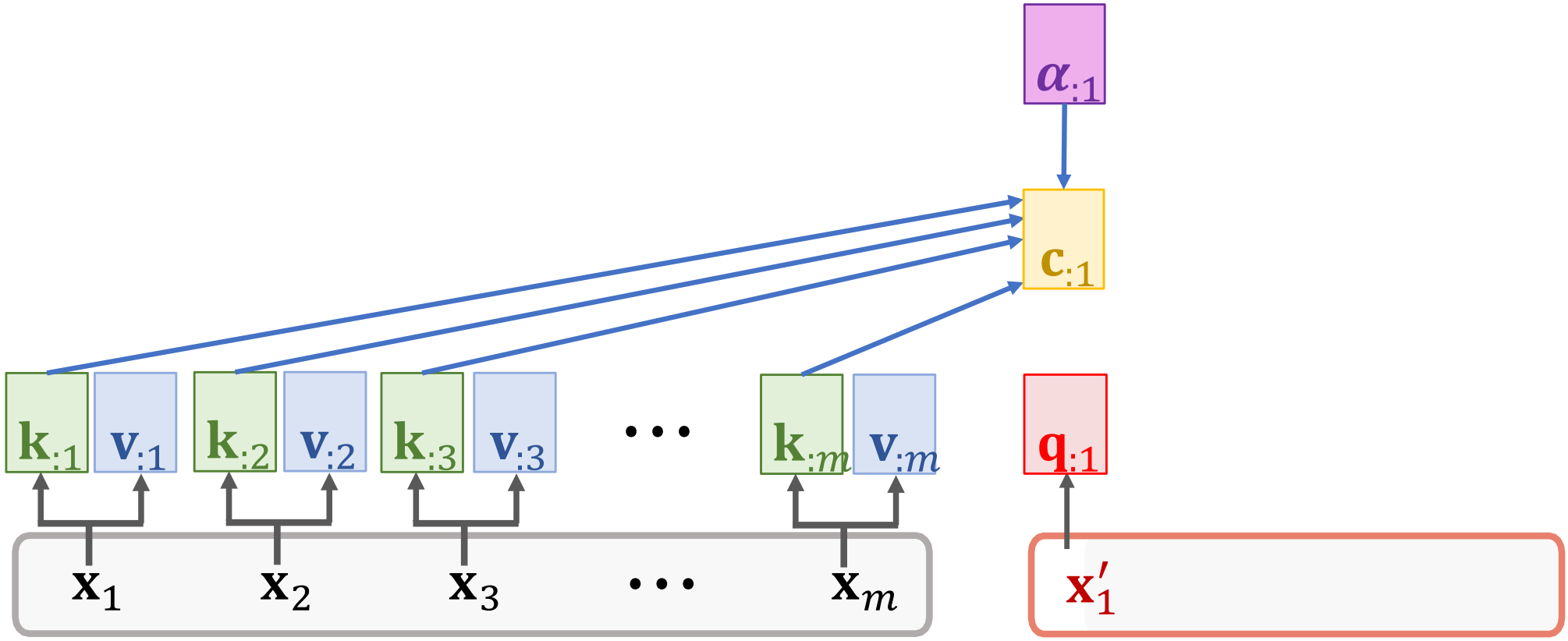
Attention w/o RNN

➢计算权重 $\boldsymbol{\alpha}_{:1} = \mathrm{Softmax}(\mathbf{K}^\top \mathbf{q}_{:1}) \in \mathbb{R}^m$



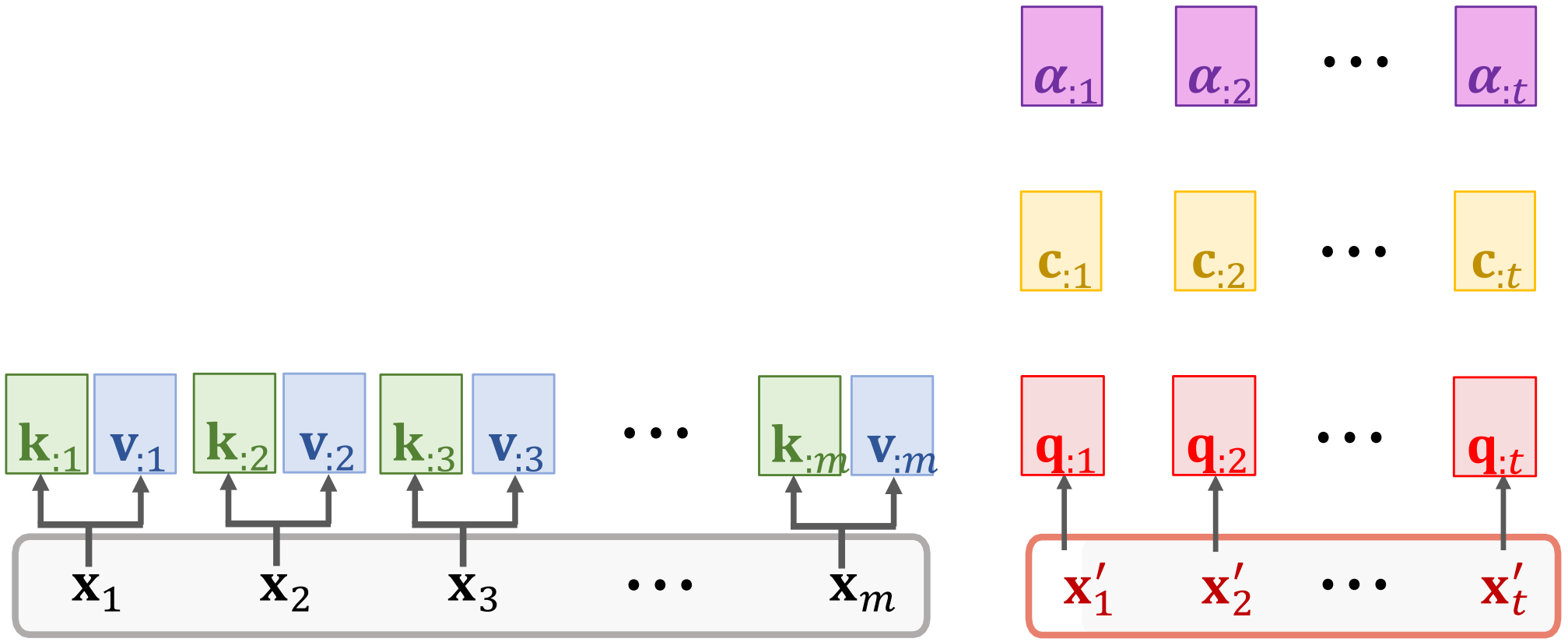
Attention w/o RNN

- 计算权重 $\boldsymbol{\alpha}_{:1} = \mathrm{Softmax}(\mathbf{K}^\top \mathbf{q}_{:1}) \in \mathbb{R}^m$
- 计算上下文向量 $\mathbf{c}_{:1} = \boldsymbol{\alpha}_{11}\mathbf{v}_{:1} + \cdots + \boldsymbol{\alpha}_{m1}\mathbf{v}_{:m} = \mathbf{V}\boldsymbol{\alpha}_{:1}$



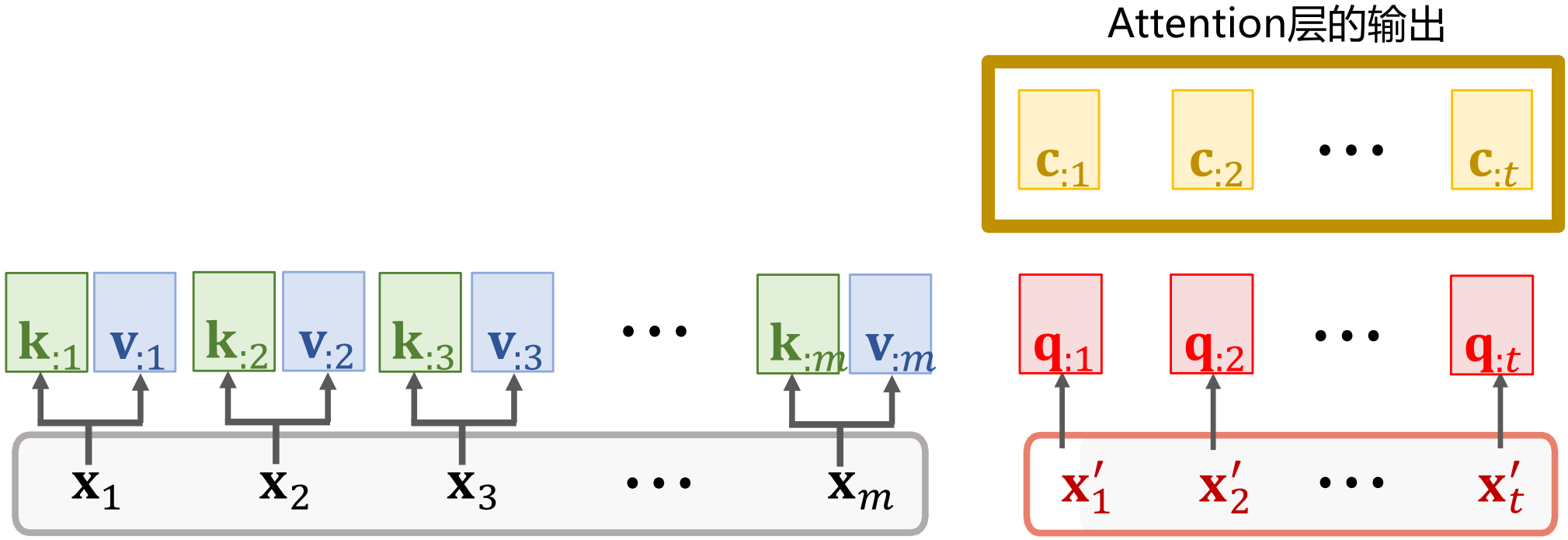
Attention w/o RNN

- 计算权重 $\boldsymbol{\alpha}_{:j} = \text{Softmax}\left(\mathbf{K}^\top \mathbf{q}_{:j}\right) \in \mathbb{R}^m$
- 计算上下文向量 $\mathbf{c}_{:j} = \boldsymbol{\alpha}_{1j}\mathbf{v}_{:1} + \cdots + \boldsymbol{\alpha}_{mj}\mathbf{v}_{:m} = \mathbf{V}\boldsymbol{\alpha}_{:j}$



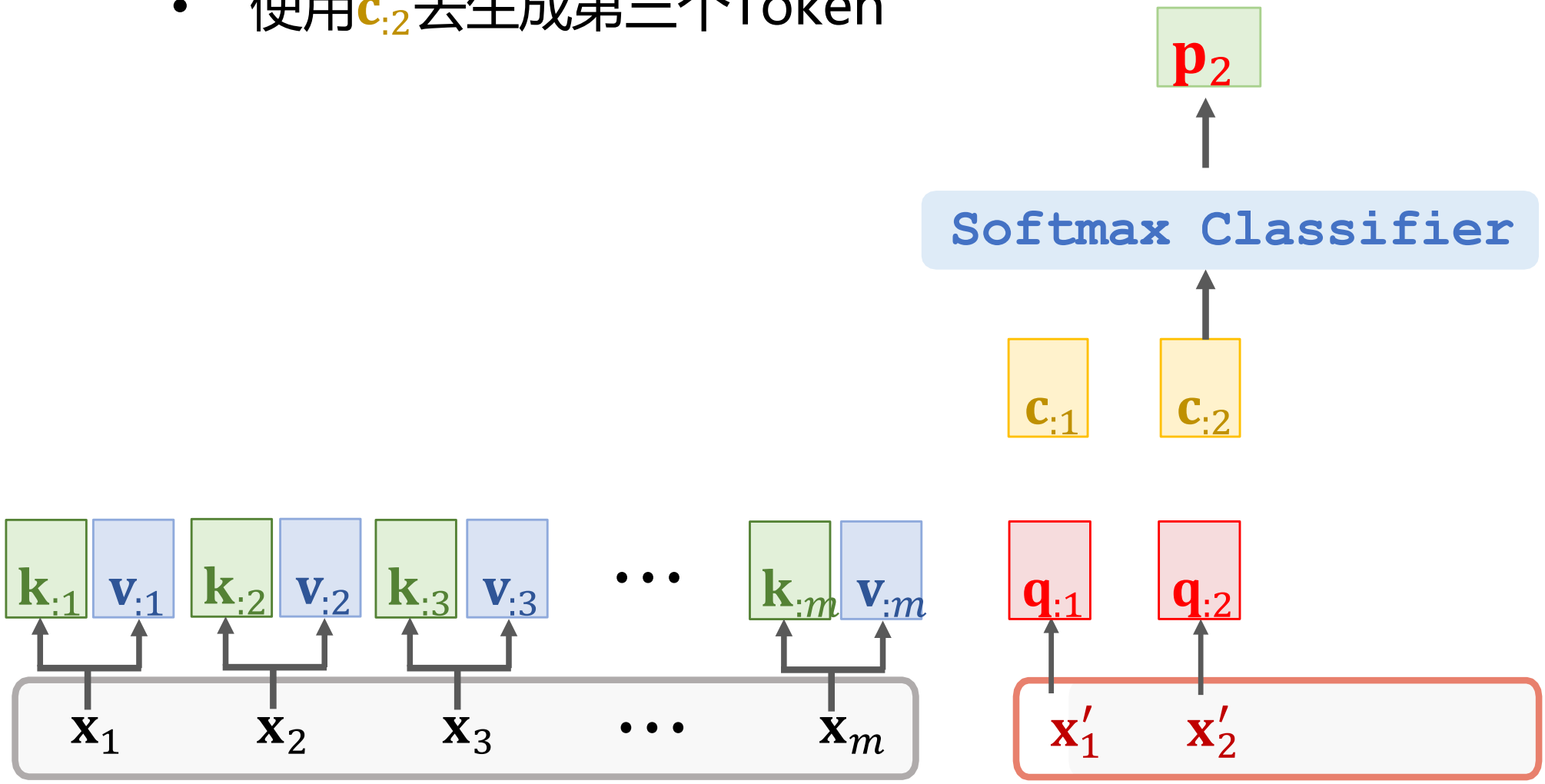
Attention w/o RNN

- Attention层的输出：$\mathbf{C} = [\mathbf{c}_{:1}, \mathbf{c}_{:2}, \mathbf{c}_{:3}, \cdots, \mathbf{c}_{:t}]$
  - 其中，$\mathbf{c}_{:j} = \mathbf{V} \cdot \mathrm{Softmax}\left(\mathbf{K}^{\top} q_{:j}\right)$
  - 因此，$\mathbf{c}_{:j}$是$\mathbf{x}'_j$和$[\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_m]$的函数



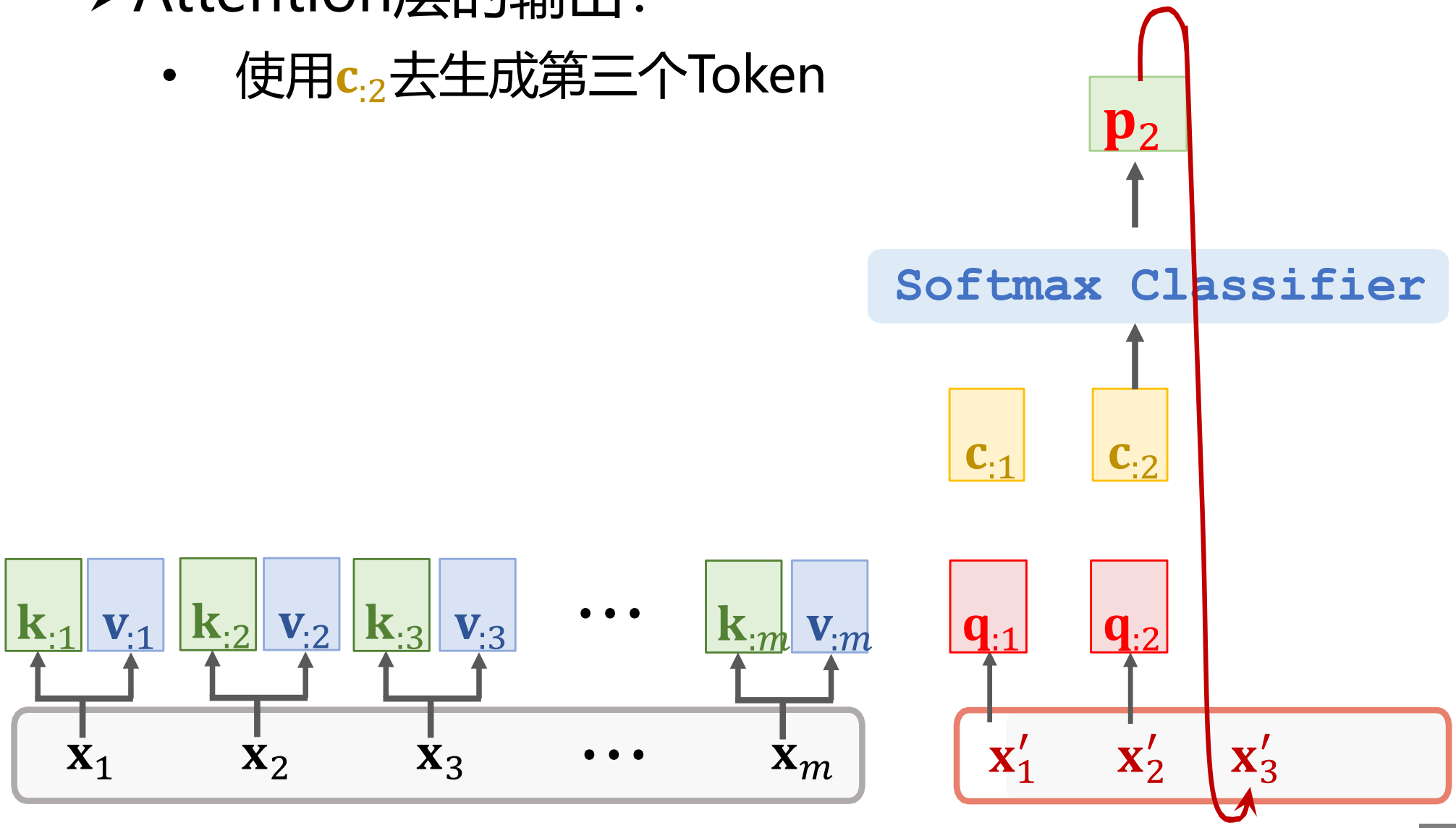
Attention w/o RNN

- 将Attention层用于机器翻译
  - 使用$\mathbf{c}_{:2}$去生成第三个Token



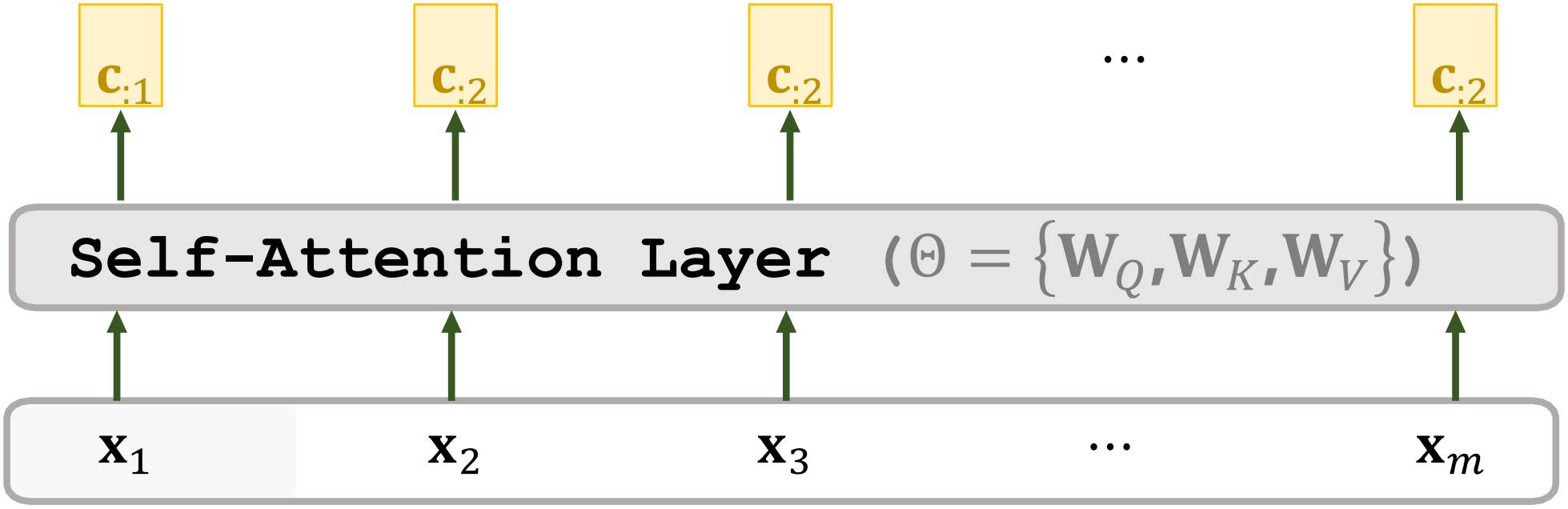
Attention w/o RNN

- Attention层的输出：
  - 使用$\mathbf{c}_{:2}$去生成第三个Token



Self-Attention Layer

- 注意力层：$\mathbf{C} = \text{Attn}(\mathbf{X}, \mathbf{X}')$，$\Theta = \{\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V\}$
- 自注意力层：$\mathbf{C} = \text{Attn}(\mathbf{X}, \mathbf{X})$，$\Theta = \{\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V\}$



Transformer的Encoder

Feedback forward network

（前馈神经网络）

Skip connection & Layer Normalization

（跳连 & 层归一化）

Self-attention & Multi-head attention

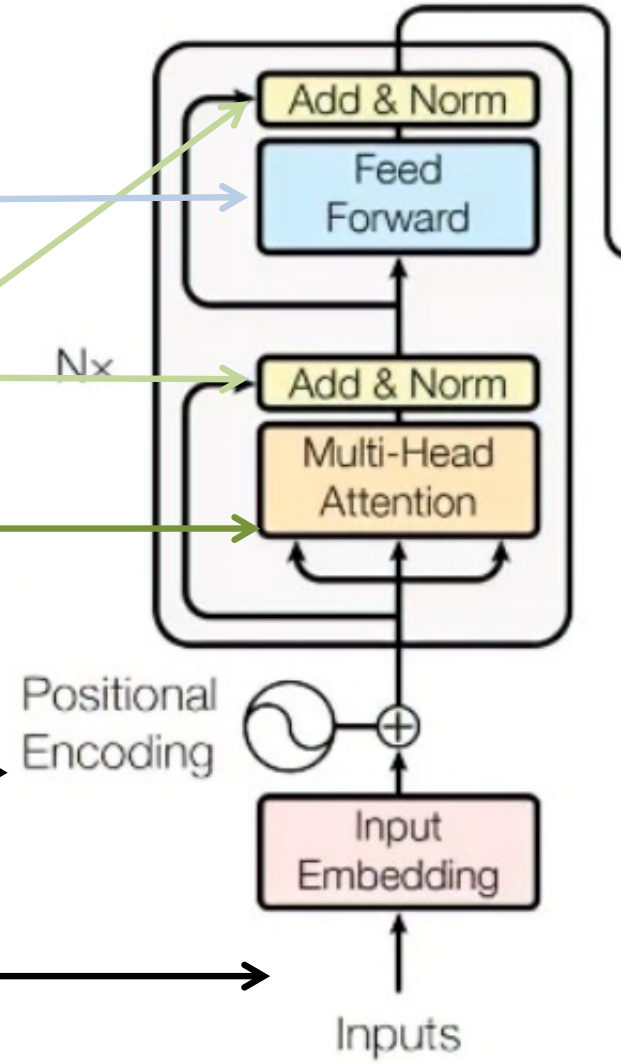
（自注意力机制 & 多头注意力机制）

Positional encoding

（位置编码）

Inputs

（输入）



谢谢大家！

